

Wheat

A Guide on Special Features of Varieties for
Different Production Conditions in India



भा.कृ.अनु.प.- भारतीय गेहूँ एवं जौ अनुसंधान संस्थान

करनाल - 132001, भारत

ICAR- Indian Institute of Wheat and Barley Research

Karnal - 132001, India



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A Guide on Special Features of Varieties for Different Production Conditions in India

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Correct Citation : Gyanendra Singh, BS Tyagi, Arun Gupta, Vineet Kumar, Vinod Tiwari, R Chatrath, R Tiwari, SK Singh, MS Saharan, RK Sharma, RK Gupta and Indu Sharma (2016). Wheat : A Guide on Special Features of Varieties for Different Production Conditions in India. Indian Institute of Wheat & Barley Research, Karnal - 132001 (India). Research Bulletin No 36. 68p

First print : January, 2016

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Indian Institute of Wheat and Barley Research, Karnal - 132001 (India)

Copies : 500

Published by: Dr. Indu Sharma,
Director,
Indian Institute of Wheat and Barley Research,
Karnal - 132001 (India)

Printed at : Shrikoshi Reprographics
121, Sector -3, Indl. Area,
Karnal - 132001
Tel. : 9812053552

Preface

Wheat is the second most important cereal crop occupying nearly 30 million hectare area in the country. This crop has been under cultivation in the Indian subcontinent and three species of wheat namely, *Triticum aestivum* (Bread wheat), *Triticum durum* (Macaroni wheat) and *Triticum dicoccum* (Emmer or Khapli) of spring types are grown during winter season. Another wheat species that used to be cultivated in the ancient past (Indus Valley civilization period) and which now has almost vanished is *T. sphaerococcum* or Indian dwarf wheat.

It was sheer coincidence that the advent of dwarf wheat and establishment of the All India Coordinated Wheat Improvement Project (AICWIP) happened concurrently in the year of 1965. This was an important milestone that brought about systematic developments in wheat research and had resulted to the real breakthrough in productivity. Since then the wheat research had made spectacular advances in yield potential, productivity and sustainability. In view of ever-growing population, changing climate, dwindling water resources, deteriorating soil health, increasing micro nutrient deficiencies and decreasing arable land, the challenge is really critical to achieve the target production. Therefore, researchers and other stake holders cannot be complacent as the challenges are myriad in the years ahead.

The choice of variety in coming time would depend on potential of genotype to yield better under changing environmental conditions, meeting requirements for tillage x genotype interaction, suitability to diversify cropping systems, management of diseases and pests and quality needs. Besides, some of the immediate steps in order to boost the wheat production levels in different agro-ecological zones of the country will include timely availability of quality seed of recently released varieties, balanced used of fertilizers, weed control measures and suitable small farm machinery.

This publication is presented to assist researchers, extension agencies and farmers in making the choice of suitable wheat variety for different zones and production conditions alongwith their special features, crop management practices and proper approach to management of diseases and pests.

The authors acknowledge the encouragement and support in bringing out this publications and hope that it will serve as a reference guide to make a wise decision for meeting the need based requirements of various stakeholders.



(Indu Sharma)

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Preamble

Wheat (*Triticum aestivum* L.) is one of the most important cereal crops of the world and occupies a notable position not only in terms of area and production but also in respect of its adaptation to a wider range of agro-climatic conditions. India is the second largest producer of wheat in the world, next only to China. Wheat contributes about 35 per cent of the total cereals production in the country. The wheat varieties developed and released in India during the last six decades, were very successful in increasing the wheat production from a mere 12.5 mt in 1964 to the historic 95.85 mt during the crop year 2013-14. The phenomenal increase in area, production and productivity of wheat established newer records with each passing year. The major goals for wheat improvement include higher yield with good grain quality, resistance to rusts, head scab and spot blotch; tolerance to abiotic stresses viz; initial and terminal heat especially during grain filling period, drought, salinity and waterlogging.

Agro-climatic zones : Wheat crop in India is grown under six diverse agro-climatic zones, wherein Indo-Gangetic Plains (IGPs) comprising the North Western Plains Zone (NWPZ) and the North Eastern Plains Zone (NEPZ) form the major wheat cultivation area constituting about 20 mha followed by the Central Zone (CZ) and the Peninsular Zone (PZ). The classification of zones is based on climatic conditions, soil types and crop duration. However, realizing the vast variations in climatic conditions in different regions of India, the ongoing programmes are tuned to address even the micro-niche as well as regional specific needs. It has been found that certain wheat genotypes that were primarily developed for rainfed conditions produce higher yield when provided one-two irrigations. Some of the promising wheat varieties namely, DBW 110, HI 1500, HI 1531, HI 8627, PBW 596 etc. have recorded production of around 35 q/ha under restricted irrigated conditions as compared to 15-20 quintal under rainfed conditions. Hence, all possible efforts are made to develop water use efficient (WUE) wheat varieties and provide appropriate wheat cultivation technology to farmers to face the shortage of water for irrigation.

Biotic stresses : The host resistance is the cheapest, most effective and environmental friendly means of managing wheat diseases. The yellow (stripe) rust disease is prevalent in northern and southern hills, north western and north eastern plains of the country. Whereas, brown (leaf) rust is common across all zones and black (stem) rust is important in central and peninsular India. Since most of the varieties recommended for NWPZ and NHZ do not carry high level of resistance, hence, chemical sprays with 0.1% propiconazole (@ 0.1 per cent), or tebuconazole or triademefon at stripe rust initiation stage is advised. Other important diseases include Karnal bunt, foliar blight, powdery mildew and loose smut. In the same way, loose smut and powdery mildew diseases of wheat are very well managed through seed treatment and spray of recommended fungicides, respectively. Nevertheless, in view of decreasing water table and increasing temperature, both termites and aphids may become prominent

threats to wheat crop in future. The incidences of termites and aphids have sometimes been reported from certain parts, which can be controlled by use of recommended insecticides.

Abiotic stresses : The major wheat growing states such as Uttar Pradesh, Punjab, Haryana, Rajasthan, Madhya Pradesh, Chhattisgarh, Bihar and Jharkhand are considered more vulnerable to rise in temperature that adversely affect both grain quality and yield. To avoid such limitations, appropriate choice of heat tolerant genotypes should be made available to the growers. In general, timely sown wheat crop is more prone to lodging due to heavy foliage causing yield losses ranging between 5-25%. In view of this, it becomes necessary to grow genotypes that possess some degree of tolerance to lodging under conditions of excess rains and heavy storms at the time of anthesis and grain filling stages.

Choice of varieties and tillage options : At present, good choice of improved varieties is available to farmers for growing under different production conditions. The selection of a suitable variety by the farmer is the one aspect of package of practices to enhance yield through matching production technology. Along with choice of recommended varieties, there is need for enhancing productivity through agronomic interventions and resource conservation technologies to save on input costs. Thus, good crop management is necessary to harness the full sustainable production potential of improved varieties and new production technologies. Wheat genotypes exhibiting better seed germinability, seedling vigour and higher root biomass during early as well as later growth stages have been found more suitable for growing under zero-tillage conditions. Similarly, for FIRBs technology, wheat varieties capable of producing more effective tillers per unit area are preferred.

Quality needs : The Indian wheat is predominantly grown and known for chapati making quality, however, promising varieties suitable for various other products like bread, biscuit, semolina, pasta, macaroni etc. have also been identified and the country has been rated as the second largest biscuit industry in the world. The demand for bread, biscuit and pasta is expected to increase in the country and therefore, there is a big scope for focussing on industries offering wheat grain value addition. *Triticum dicoccum* has already established itself as a good cereal crop for diabetic patients. Packaged atta with different proportion of multi grains in wheat flour makes it more suitable for urban consumers.

This publication highlights the special features of promising varieties suitable for different zones and production conditions alongwith package of practices, crop management options to enable wheat growers to choose a variety as per their specific situations and local demand so as to increase production and productivity and fulfil quality needs for changing food habits.

Northern Hill Zone

Varieties at a glance

Production condition	Variety
Early sown, Rainfed	HS 542, HPW 251, VL 829
Timely sown, Rainfed /Irrigated	HPW 349, HS 507, VL 907, VL 804
Late sown, Restricted irrigated	HS 490, VL 892, HS 420
High altitude areas	VL 832, HS 375

Package of Practices

Condition	Sowing time	Seed rate (kg/ha)	Fertilizer doses and time of application
Early sown, Rainfed	1 st fortnight of October	100	60:30:20 kg NPK/ha at the time of sowing
Timely sown, Rainfed	2 nd fortnight of October	100	60:30:20 kg NPK/ha at the time of sowing
Timely sown, Irrigated	1 st fortnight of November	100	120:60:40 kg NPK /ha(1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Late sown, Restricted irrigated	1 st fortnight of December	125	90:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
High altitude areas	2 nd fortnight of April & May	100	60:30:20 kg NPK/ha at the time of sowing

HS 542

Zone: Northern Hill Zone

Production condition: Early sown, Rainfed

Parentage	MILAN/KAUZ//PRINIA/3/BABAX
Year of release	2015
Not. no. & date	268(E), 28-01-2015
Developed by	IARI, RS, Shimla

Morphological features

Plant height (cm)	91
Days to maturity (days)	192
1000-grain weight (g)	44
Average yield (q/ha)	32.9
Potential yield (q/ha)	49.3

Quality parameters

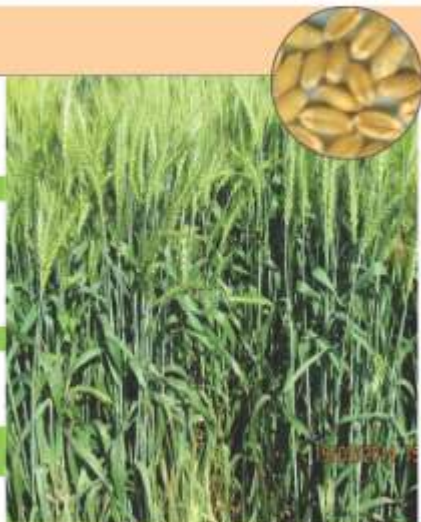
Chapati score (max. 10)	7.88
Bread loaf volume (cc)	522
Biscuit spread factor (max. 10)	7.57

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer doses

Grain yield under N60 kg/ha	34.7
Grain yield under N80	37.7

Special Features: High yielding under early sown rainfed condition, responsive to higher nitrogen levels, possesses resistance against yellow & brown rust and good grain for chapati making.



HPW 251

Zone: Northern Hill Zone

Production condition: Early sown, Rainfed

Parentage	WW 24/LEHMI
Year of release	2008
Notification year	1108(E), 08-05-2008
Developed by	CSKHPKV, Palampur

Morphological features

Plant height (cm)	89
Days to maturity (days)	177
1000-grain weight (g)	40
Average yield (q/ha)	34.43
Potential yield (q/ha)	40.1

Quality Characters

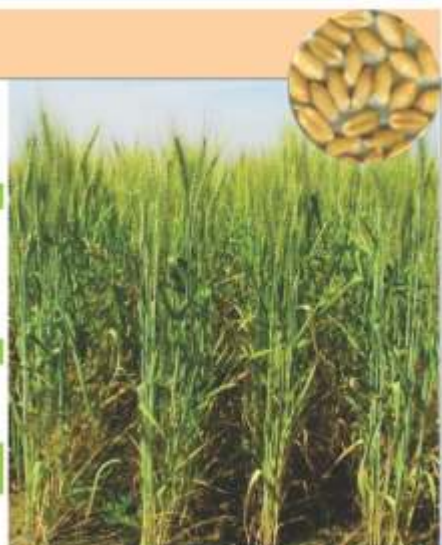
Chapati score (max 10)	7.24
Bread loaf volume (cc)	535
Protein (%)	10.93

Agronomical practices –

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	30.4
Grain yield under N80	34.0

Special Features: High grain yield under early sown rainfed conditions, possesses diversity for seedling resistance for leaf rust and is moderately resistant to Karnal bunt. Good for chapati making and has high iron, copper and manganese contents.



VL 829

Zone: Northern Hills Zone

Production condition: Early sown, Rainfed

Parentage IBWSN 149/CPAN 2099

Year of release 2002

Notification No. 283(E), 12-03-2003

Developed by VPKAS, Almora

Morphological features

Plant height (cm)	93
Days to maturity (days)	208
1000-grain weight (g)	45
Average yield (q/ha)	29.1
Potential yield (q/ha)	59.8

Quality parameters

Protein content (%)	11.36
Sedimentation value	45.3
Hectolitre weight (g)	75

Agronomic practices

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	25.1
Grain yield under N80	35.7



Special Features: Dual purpose (green fodder cum grain) wheat variety for limited irrigation condition. Resistant to brown rust and responsive to enhanced fertilizer doses under rainfed condition. High protein content and also good hectolitre weight.

HPW 349

Zone: Northern Hill Zone

Production condition: Timely sown, Rainfed/Irrigated

Parentage OASIS/KAUZ//4*BCN/3/PASTOR/4/K

AUZ*2/YACO//KAUZ

Year of release 2013

Notification No. 952(E), 10-04-2013

Developed by CSKHPKV, Palampur

Morphological features

Plant height (cm)	86 (RF), 102 (IR)
Days to maturity (days)	175 (RF), 171 (IR)
1000-grain weight (g)	37.3 (RF), 42.3 (IR)
Average yield (q/ha)	25.9 (RF), 47.0 (IR)
Potential yield (q/ha)	42.1 (RF), 61.4 (IR)

Quality parameters

Chapati score (max 10)	7.74 (RF), 7.81 (IR)
Bread loaf volume (cc)	555 (RF), 550 (IR)
Biscuit spread factor (max 10)	7.52 (RF), 7.89 (IR)

Agronomic practices

Grain yield (q/ha) under enhanced fertilizer dose (RF)

Grain yield (q/ha) under normal and / late planting (IR)

Grain yield under N60	26.3 (RF)	Grain yield under normal sown	42.8 (IR)
Grain yield under N80	27.2 (RF)	Grain yield under late sown	32.9 (IR)



Special Features: High yield potential for rainfed as well as irrigated conditions. Responsive to higher doses of nitrogen fertilizer. Resistant to yellow & brown rusts as evident by diversity for seedling resistance genes. The variety is good in chapati making and also better nutrition quality.

HS 507

Zone: Northern Hill Zone

Production condition: Timely sown, Rainfed/Irrigated

Parentage	KAUZ/MYNA/VUL/BUC/FLK/4/MILAN
Year of release	2011
Notification No.	622(E), 25-03-2011
Developed by	IARI, RS, Shimla

Morphological features

Plant height (cm)	77(RF), 95(IR)
Days to maturity (days)	179(RF), 165(IR)
1000-grain weight (g)	36(RF), 41(IR)
Average yield (q/ha)	26.6 (RF), 46.8(IR)
Potential yield (q/ha)	54.3 (RF), 60.1 (IR)

Quality parameters

Chapati score (max 10)	7.53(RF), 7.45(IR)
Bread loaf volume (cc)	563(RF), 560(IR)
Biscuit spread factor (max 10)	7.92(RF), 7.82 (IR)

Agronomic practices-

Grain yield (q/ha) under normal/ late planting (IR)

Grain yield (q/ha) under enhanced fertilizer dose (RF)

Grain yield under normal sown	40.4(IR)	Grain yield under N60	26.4 (RF)
Grain yield under late sown	32.3(IR)	Grain yield under N80	28.1(RF)

Special Features: Substantial yield superiority under both rainfed and irrigated conditions. This variety has high degree of resistance to leaf and stripe rusts. Good for chapati, bread and biscuit making.



VL 907

Zone: Northern Hill Zone

Production condition: Timely sown, Rainfed/Irrigated

Parentage	DYBR 1982-83/842 ABVD 50/VW
	9365//PBW 343
Year of release	2010
Notification No.	733(E), 01-04-2010
Developed by	VPKAS, Almora

Morphological features

Plant height (cm)	82(RF), 97(IR)
Days to maturity (days)	180(RF), 167(IR)
1000-grain weight (g)	39.2(RF), 41.0 (IR)
Average yield (q/ha)	27.9(RF), 44.3(IR)
Potential yield (q/ha)	52.5 (RF), 56.9(IR)

Quality parameters

Chapati score (max 10)	7.364(RF), 7.58(IR)
Bread loaf volume (cc)	548(RF), 543(IR)
Biscuit spread factor (max 10)	5.91(RF), 6.76(IR)

Agronomic practices-

Grain yield under enhanced fertilizer dose (RF)

Grain yield (q/ha) under normal/ late planting (IR)

Grain yield under N40	28.7 (RF)	Grain yield under normal sown	52.5(IR)
Grain yield under N60	31.9(RF)	Grain yield under late sown	41.8 (IR)

Special Features: Yield superiority under rainfed and irrigated conditions. This genotype is good source of micro elements iron, zinc, copper and manganese. Resistant to yellow and brown rusts.



VL 804

Zone: Northern Hill Zone

Production condition: Timely sown, Rainfed/Irrigated

Parentage CPAN 3018/CPAN 3004//PBW 65

Year of release 2002

Notification No. 937(E), 04-09-2002

Developed by VPKAS, Almora

Morphological features

Plant height (cm)	87(RF), 102(IR)
Days to maturity (days)	178(RF), 164(IR)
1000-grain weight (g)	36.1(RF), 37.2(IR)
Average yield (q/ha)	25.7(RF), 41.3 (IR)
Potential yield (q/ha)	43.1(RF), 54.7 (IR)

Quality parameters

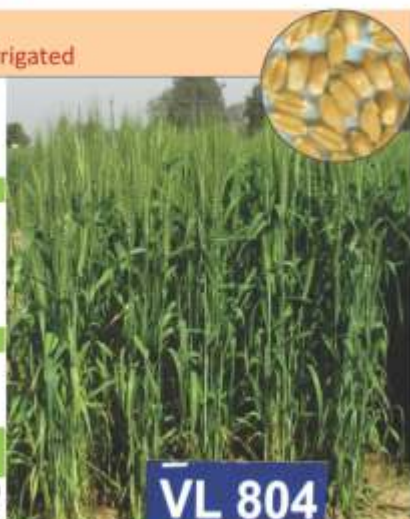
Chapati score (max 10)	7.69(RF), 7.73(IR)
Bread loaf volume (cc)	545(RF), 543(IR)
Biscuit spread factor (max 10)	7.54(RF), 7.70(IR)

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose (RF/IR)

Grain yield under (N60-RF; N 120- IR)	20.4(RF), 42.0(IR)
Grain yield under (N90-RF; N 150- IR)	26.9(RF), 44.4(IR)

Special Features: Yield superiority under both irrigated and rainfed situations, Resistant to brown rust, loose smut. Responsive to fertilizer applications with varying moisture regimes. Good for chapatti and bread making quality.



HS 490

Zone: Northern Hill Zone

Production condition: Late sown, Restricted irrigated

Parentage HS 364/HPW 114//HS 240/HS 346

Year of release 2009

Notification No. 449(E), 11-02-2009

Developed by IARI/RS, Shimla

Morphological features

Plant height (cm)	82
Days to maturity (days)	150
1000-grain weight (g)	44.3
Average yield (q/ha)	31.1
Potential yield (q/ha)	49.7

Quality parameters

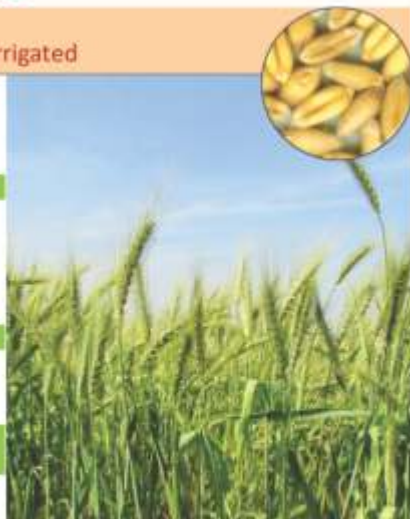
Chapati score (max 10)	7.28
Bread loaf volume (cc)	557
Biscuit spread factor (max 10)	8.72

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	36.6
Grain yield under very late sown	32.9

Special Features: High yield and wider adaptation for cultivation under late and very late sown conditions. This variety also has seedling resistance against 46S119 race of yellow rust and resistant to Karnal bunt. This variety has high spread factor, low sedimentation value and thus very good for biscuit quality.



VL 892

Zone: Northern Hill Zone

Production condition: Late sown, Restricted irrigated

Parentage	WH 542/PBW 226
Year of release	2008
Notification No.	1108(E), 08-05-2008
Developed by	VPKAS, Almora

Morphological features

Plant height (cm)	79
Days to maturity (days)	143
1000-grain weight (g)	36
Average yield (q/ha)	37.6
Potential yield (q/ha)	59.0

Quality parameters

Chapati score (max 10)	6.92
Bread loaf volume (cc)	550
Biscuit spread factor (max 10)	6.97

Agronomic practices -

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	36.8
Grain yield under late sown	37.9



Special Features: VL 892 has shown good potential for late restricted irrigation conditions of NHZ. This variety has also shown resistance against prevalent races of brown rust. It has high zinc, copper and manganese and will in nutritional security.

HS 420

Zone: Northern Hill Zone

Production condition: Late sown, Restricted irrigated

Parentage	KAJ 3302//CMH 73 A497/3*CNO 79
Year of release	2003
Notification No.	283(E), 12-03-2003
Developed by	IARI-RS, Shimla

Morphological features

Plant height (cm)	75
Days to maturity (days)	143
1000-grain weight (g)	35
Average yield (q/ha)	26.7
Potential yield (q/ha)	44.7

Quality parameters

Protein content (%)	11.4
Sedimentation value	59.4
Hectolitre weight (g)	75.5

Agronomic practices -

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	26.5
Grain yield under N80	28.9



Special Features: Higher yield under recommended dose of nitrogen and also showed suitability for enhanced levels of fertilizers. High protein content and high sedimentation value.

VL 832

Zone: Northern Hill Zone

Production condition: High altitude areas

Parentage	PBW 65/ CPAN 3031
Year of release	2004
Notification No.	161(E), 04-02-2004
Developed by	VPKAS, Almora

Morphological features

Plant height (cm)	96
Days to maturity (days)	183
1000-grain weight (g)	45.0
Average yield (q/ha)	26.2
Potential yield (q/ha)	35.2

Quality parameters

Chapati score (max 10)	-
Bread loaf volume (cc)	-
Biscuit spread factor (max 10)	-

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	23.0
Grain yield under N80	29.5

Special Features: Goodyield potential adaptability to nitrogen levels. The variety has resistance to brown rust and loose smut.



HS 375

Zone: Northern Hill Zone

Production condition: High altitude areas

Parentage	BB/G11/CJ71/3/TAEST// KAL/BB
Year of release	2003
Notification No.	283(E), 12-03-2003
Developed by	IARI-RS, Shimla

Morphological features

Plant height (cm)	85
Days to maturity (days)	113
1000-grain weight (g)	50.1
Average yield (q/ha)	26.6
Potential yield (q/ha)	49.8

Quality parameters

Protein content (%)	10.26
Sedimentation value	55.5
Hectolitre weight (g)	78.9

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	-
Grain yield under late sown	-

Special Features: HS 375 has good yielding ability under very high altitude areas. The genotype has shown resistance against leaf rusts.



North Western Plains Zone

Varieties at a glance

Condition	Variety
Timely sown, Irrigated	Bread Wheat : DBW 88, HD 3086, WH 1105, HD 2967, DPW 621-50, PBW 550, DBW 17 Durum Wheat : WHD 943, PDW 314, PDW 291
Late sown, Irrigated	DBW 90, WH 1124, DBW 71, HD 3059, PBW 590, WH 1021, DBW 16, RAJ 3765
Timely sown, Rainfed/ Restricted irrigated	HD 3043, WH 1080, PBW 396, PBW 644

Package of Practices

Condition	Sowing time	Seed rate	Fertilizer doses and time of application
Timely sown Irrigated	2 nd -3 rd week of November	100 kg/ha	150:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Late sown Irrigated	3 rd – 4 th week of December	120 kg/ha	120:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Timely sown Restricted irrigated	4 th week of October- 1 st week of November	100 kg/ha	90:60:40 kg NPK/ha at the time of sowing

DBW 88

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	KAUZ//ALTAR84/AOS/3/MILAN/ KAUZ/4/HUITES
Year of release	2014
Notification No.	244(E), 24-01-2014
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	97
Days to maturity (days)	143
1000-grain weight (g)	38
Average yield (q/ha)	54.2
Potential yield (q/ha)	69.9

Quality parameters

Chapati score (max 10)	7.66
Bread loaf volume (cc)	565
Biscuit spread factor (max 10)	7.06

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	54.3
Grain yield under late sown	44.7



Special Features: DBW 88 has higher protein content, resistant to yellow and brown rust and elasticity for sowing time thereby suitable for varying planting dates.

HD 3086

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	DBW 14/ HD 2733 // HUW 468
Year of release	2014
Notification No.	244(E), 24-01-2014
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	98
Days to maturity (days)	143
1000-grain weight (g)	39
Average yield (q/ha)	54.6
Potential yield (q/ha)	71.1

Quality parameters

Chapati score (max 10)	7.65
Bread loaf volume (cc)	565
Biscuit spread factor (max 10)	6.85

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	53.5
Grain yield under late sown	43.5



Special Features: This variety has higher yield, resistant to yellow & brown rust.

WH 1105

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	MILAN/S87230//BABAX
Year of release	2013
Notification No.	952(E), 10-04-2013
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	99
Days to maturity (days)	142
1000-grain weight (g)	41.1
Average yield (q/ha)	52.5
Potential yield (q/ha)	71.6

Quality parameters

Chapati score (max 10)	7.60
Bread loaf volume (cc)	548
Biscuit spread factor (max 10)	7.80

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	58.3
Grain yield under late sown	44.1

Special Features: High yield and resistant to yellow rust, leaf blight and powdery mildew diseases, possesses good chapati making quality and has tolerance to terminal heat stress.



HD 2967

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	ALD/COC//URES/HD2160M/HD2278
Year of release	2011
Notification No.	2326(E), 10-10-2011
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	98
Days to maturity (days)	143
1000-grain weight (g)	42.1
Average yield (q/ha)	50.4
Potential yield (q/ha)	66.1

Quality parameters

Chapati score (max 10)	7.64
Bread loaf volume (cc)	573
Biscuit spread factor (max 10)	6.39

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	54.8
Grain yield under late sown	43.5

Special Features: Recommended for two mega zones (NWPZ & NEPZ). This genotype has adult plant resistance to yellow and brown rusts. It has high zinc, copper and iron thereby securing nutritional security.



DPW 621-50

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	KAUZ//ALTAR84/AOS/3/MILAN/ KAUZ/4/HUITES
Year of release	2011
Notification No.	1661(E), 27-01-2011
Developed by	IIWBR, Karnal; PAU, Ludhiana

Morphological features

Plant height (cm)	97
Days to maturity (days)	144
1000-grain weight (g)	38
Average yield (q/ha)	51.3
Potential yield (q/ha)	69.8

Quality parameters

Chapati score (max 10)	7.69
Bread loaf volume (cc)	582
Biscuit spread factor (max 10)	6.98

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	51.3
Grain yield under late sown	41.2



Special Features: Resistant to yellow and leaf rusts. It has good loaf volume, chapati quality along with better nutritional values.

PBW 550

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	WH 594/RAJ 3858//W 485
Year of release	2008
Notification No.	72(E), 10-01-2008
Developed by	PAU, Ludhiana

Morphological features

Plant height (cm)	86
Days to maturity (days)	135
1000-grain weight (g)	39.1
Average yield (q/ha)	47.7
Potential yield (q/ha)	62.4

Quality parameters

Chapati score (max 10)	7.33
Bread loaf volume (cc)	549
Biscuit spread factor (max 10)	6.36

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	44.9
Grain yield under late sown	38.9



Special Features: High yielding genotype with adaptability to varying dates of sowing. The variety possesses resistance to leaf rust. It is good for chapati making quality and has tolerance to terminal heat.

DBW 17

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	CMH79A.95/3*CNO 79//RAJ3777
Year of release	2007
Notification No.	122(E), 16-02-2007
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	86
Days to maturity (days)	134
1000-grain weight (g)	37
Average yield (q/ha)	49.0
Potential yield (q/ha)	63.0

Quality parameters

Chapati score (max 10)	7.95
Bread loaf volume (cc)	565
Biscuit spread factor (max 10)	7.22

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	49.0
Grain yield under late sown	42.1

Special Features: DBW 17 has shown high yielding ability under timely and late sown conditions. Tolerant to terminal heat stress and has faster grain filling. It has resistance to yellow rust, pathotypes 78S84 and leaf blight diseases. It has good grain appearance, hectolitre weight and chapati quality.



WHD 943 (Durum)

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	GLARE/PLATA16//AJAIA-3/SILVER 16
Year of release	2011
Notification No.	633(E), 25-03-2011
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	89
Days to maturity (days)	144
1000-grain weight (g)	44.1
Average yield (q/ha)	48.1
Potential yield (q/ha)	63.9

Quality parameters

Yellow pigment (ppm)	6.8
Grain hardness index	82
Protein content (%)	12.5

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	48.9
Grain yield under late sown	40.3

Special Features: WHD 943 showed high resistance to yellow rust and better adaptability to normal and late sown conditions. This variety has high protein, high nutrient uptake and good pasta making quality.



PDW 314 (Durum)

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	AJAI 12/F3 LOCAL (SEL. ETHIO.135.85)//PLATA 13/3/SOMAT 3/4/SOOTY9/RASCON 37
Year of release	2010
Notification No.	733(E), 01-04-2010
Developed by	PAU, Ludhiana

Morphological features

Plant height (cm)	89
Days to maturity (days)	146
1000-grain weight (g)	41
Average yield (q/ha)	50.3
Potential yield (q/ha)	71.2

Quality parameters

Yellow pigment (ppm)	5.1
Protein content (%)	12.3
Sedimentation value	35.7

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	51.9
Grain yield under late sown	45.6

Special Features: PDW 314 has shown wider adaptability for timely and late sowing. It has resistance to rust and leaf blight diseases. This genotype has better sedimentation value, protein content and good pasta making quality with higher level of nutritional attributes.



PDW 291 (Durum)

Zone: North Western Plains Zone

Production condition: Timely sown, Irrigated

Parentage	BOOMER 21/ MOJO2
Year of release	2005
Notification No.	122(E), 02-02-2005
Developed by	PAU, Ludhiana

Morphological features

Plant height (cm)	86
Days to maturity (days)	143
1000-grain weight (g)	43.1
Average yield (q/ha)	47.6
Potential yield (q/ha)	58.1

Quality parameters

Protein content (%)	11.7
Yellow pigment (ppm)	5.4
Hectolitre weight (g)	80.6

Agronomic practices-

Grain yield (q/ha) under enhanced fertilization

Grain yield under N120	39.0
Grain yield under N150	41.6

Special Features: PDW 291 showed better response to high doses of nitrogen and also less deduction under late sowing. It possesses high level of resistance against rust, Karnal bunt and loose smut. This is a semi dwarf genotype tolerant to lodging and has bold lustrous grains.



DBW 90

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage HUW 468/ WH 730

Year of release 2014

Notification No. 244(E), 24-01-2014

Developed by IIWBR, Karnal

Morphological features

Plant height (cm)	91
Days to maturity (days)	121
1000-grain weight (g)	35
Average yield (q/ha)	42.7
Potential yield (q/ha)	66.6

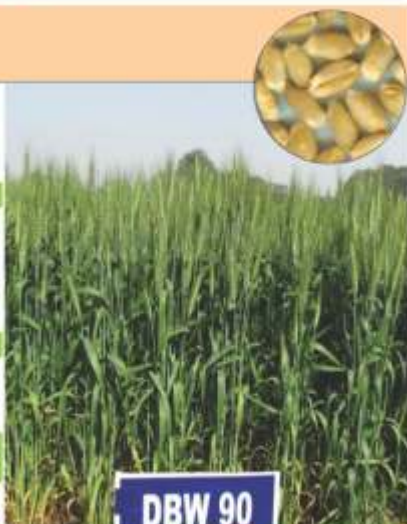
Quality parameters

Chapati score (max 10)	7.60
Bread loaf volume (cc)	561
Biscuit spread factor (max 10)	7.71

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	45.0
Grain yield under very late sown	35.1



Special Features: Variety has yield superiority under late and very late sowings as evident by minimum reduction under very late sown condition and has good degree of heat tolerance. The differential gene combination for durable resistance to stripe and leaf rusts might be helpful in minimizing disease losses. This variety is good for chapati, bread and biscuit and has better nutritional quality.

WH 1124

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage MUNIA/CHTO/AMSEL

Year of release 2014

Notification No. 1919 (E), 30-07-2014

Developed by CCSHAU, Hisar

Morphological features

Plant height (cm)	91
Days to maturity (days)	121
1000-grain weight (g)	35.1
Average yield (q/ha)	42.7
Potential yield (q/ha)	56.1

Quality parameters

Chapati score (max 10)	7.46
Bread loaf volume (cc)	569
Biscuit spread factor (max 10)	7.16

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	45.0
Grain yield under very late sown	36.4



Special Features: WH 1124 has wider adaptability, showed least yield fluctuation under different planting regimes. This variety has high degree of resistance to yellow and brown rust. This is a suitable genotype for chapati and has comparatively better nutritional quality in terms of iron and zinc.

DBW 71

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage	PRINIA/UP2425
Year of release	2013
Notification No.	2817 (E), 19-09-2013
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	90
Days to maturity (days)	119
1000-grain weight (g)	37.1
Average yield (q/ha)	42.7
Potential yield (q/ha)	68.9

Quality parameters

Chapati score (max 10)	7.49
Bread loaf volume (cc)	559
Biscuit spread factor (max 10)	7.73

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	42.9
Grain yield under very late sown	35.5



Special Features: The high yield potential of DBW 71 under irrigated late sown condition of NWPZ possessing disease resistance and heat tolerance. The genotype has high protein, chapati score, yellow pigment and micro nutrients.

HD 3059

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage	KAUZ//ALTAR84/AOS/3/MILAN/KAUZ/ 4/HUITES
Year of release	2013
Notification No.	312(E), 01-02-2013
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	93
Days to maturity (days)	121
1000-grain weight (g)	36.0
Average yield (q/ha)	42.5
Potential yield (q/ha)	59.4

Quality parameters

Chapati score (max 10)	7.58
Bread loaf volume (cc)	564
Biscuit spread factor (max 10)	7.81

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	43.9
Grain yield under late sown	35.1



Special Features: HD 3059 is an early maturing, medium dwarf genotype having high yield potential under late and even very late sown conditions. It has displayed thermo-tolerance as reflected by minimum yield loss under very late sown conditions. This variety has sown resistance against rusts. It has high protein, high sedimentation value and good bread, biscuit and chapati making qualities.

PBW 590

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage	WH 594/RAJ3814//W 485
Year of release	2009
Notification No.	454(E), 11-02-2009
Developed by	PAU, Ludhiana

Morphological features

Plant height (cm)	79
Days to maturity (days)	121
1000-grain weight (g)	36.1
Average yield (q/ha)	42.2
Potential yield (q/ha)	7.32

Quality parameters

Chapati score (max 10)	7.52
Bread loaf volume (cc)	578
Biscuit spread factor (max 10)	6.17

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	41.4
Grain yield under very late sown	32.6



Special Features: PBW 590 has high yield potential under late and very late sowing conditions indicating flexibility for sowing time and tolerance to thermal heat stress. This genotype possesses resistance to leaf rust and has high protein content along with chapati making quality.

WH 1021

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage	NYOT95/SONAK
Year of release	2008
Notification No.	1108(E), 08-05-2008
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	94
Days to maturity (days)	121
1000-grain weight (g)	36
Average yield (q/ha)	39.1
Potential yield (q/ha)	62.5

Quality parameters

Chapati score (max 10)	7.09
Bread loaf volume (cc)	572
Biscuit spread factor (max 10)	7.82

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	37.3
Grain yield under very late sown	28.2



Special Features: Bread wheat variety WH 1021 has shown high yield potential, brown rust resistance with divergence and excellent grain quality.

DBW 16

Zone: North Western Plains Zone

Production condition: Late sown, Irrigated

Parentage	RAJ 3765/WR 484//HUW 468
Year of release	2006
Notification No.	599(E), 25-04-2006
Developed by	IWBR, Karnal

Morphological features

Plant height (cm)	87
Days to maturity (days)	120
1000-grain weight (g)	35.1
Average yield (q/ha)	38.2
Potential yield (q/ha)	56.4

Quality parameters

Chapati score (max 10)	7.18
Bread loaf volume (cc)	538
Biscuit spread factor (max 10)	6.68

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	41.8
Grain yield under very late sown	29.4

Special Features: DBW 16 has shown superior and stable yielding ability under late sown conditions. It has shown high level of resistance against the prevalent pathotypes of stripe rust. This variety also has good chapati and bread, biscuit score hence suitable for multiple end products.



RAJ 3765

Zone: North Western Plains Zone

Production condition: Irrigated, late/very late sown

Parentage	HD 2402/VL 639
Year of release	1996
Notification No.	SO1(E)
Developed by	RAU, Durgapura

Morphological features

Plant height (cm)	90
Days to maturity (days)	134
1000-grain weight (g)	39.0
Average yield (q/ha)	36.5
Potential yield (q/ha)	43.8

Quality parameters

Chapati score (max 10)	7.63
Bread loaf volume (cc)	534
Biscuit spread factor (max 10)	7.56

Agronomical features-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	36.4
Grain yield under very late sown	28.3

Special Features: Raj 3765 has light green, non-waxy leaves that showed tolerance against temperature stress. It has also shown field resistance to rusts. Raj 3765 is potential genotype for late and very late sown condition in NWPZ.



HD 3043

Zone: North Western Plains Zone

Production condition: Timely sown, Rainfed/ Restricted irrigated

Parentage	PJN/BOW//OPATA*2/3/CROC_1/ <i>Ae.squarrosa</i> (224)//OPATA
Year of release	2012
Notification No.	456(E), 16-03-2012
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	105
Days to maturity (days)	144
1000-grain weight (g)	39.8
Average yield (q/ha)	42.8
Potential yield (q/ha)	50.2

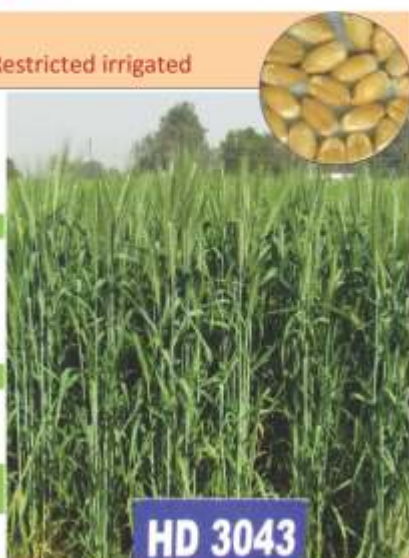
Quality parameters

Chapati score (max 10)	7.50
Bread loaf volume (cc)	546
Biscuit spread factor (max 10)	7.13

Agronomic practices-

Grain yield (q/ha) under different irrigation levels

Grain yield under one Irrigation	39.1
Grain yield under two irrigation	42.9



Special Features: This has high yield potential under limited irrigations and thus suitable for restricted irrigation conditions. HD 3043 has shown good level of resistance against leaf & stripe rust as well as adult plant resistance due to minor genes and therefore possessive durable resistance. Variety is good for chapati score and has nutrition quality.

WH 1080

Zone: North Western Plains Zone

Production condition: Timely sown, Rainfed/ Restricted irrigated

Parentage	21 st SAWSN 151
Year of release	2011
Notification No.	1661(E), 27-01-2011
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	86
Days to maturity (days)	151
1000-grain weight (g)	39
Average yield (q/ha)	30.8
Potential yield (q/ha)	44.4

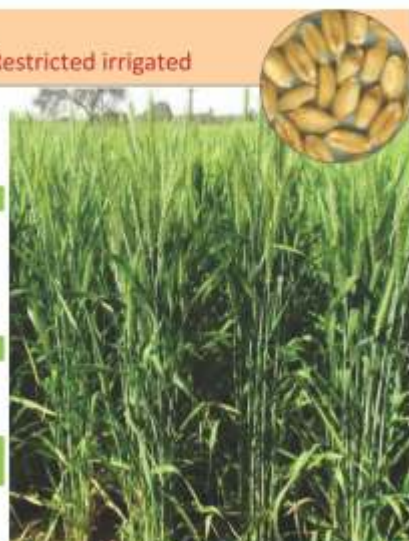
Quality parameters

Chapati score (max 10)	7.71
Bread loaf volume (cc)	584
Biscuit spread factor (max 10)	7.26

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	27.8
Grain yield under N80	29.7



Special Features: WH 1080 is suitable for rainfed timely sown and low fertility condition but has responsiveness to higher levels of nitrogen fertilizer. Genotype has shown resistance to leaf rusts, leaf blight and flag smut. Variety has high chapati score and better nutritional quality for iron and copper.

PBW 396

Zone: North Western Plains Zone

Production condition: Timely sown, Rainfed/ Restricted irrigated

Parentage	CNO67/MFD //MON"S"/3/SERI
Year of release	2000
Notification No.	340(E), 03-04-2000
Developed by	PAU, Ludhiana

Morphological features

Plant height (cm)	93
Days to maturity (days)	148
1000-grain weight (g)	38.0
Average yield (q/ha)	33.5
Potential yield (q/ha)	39.4

Quality parameters

Chapati score (max 10)	7.72(RF), 7.63(RI)
Bread loaf volume (cc)	533(RF), 548(RI)
Biscuit spread factor (max 10)	7.21(RF), 7.18(RI)

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	25.9
Grain yield under late sown	23.7



Special Features: PBW 396 is a medium dwarf genotype for rainfed areas of NWPZ and has shown resistance to yellow and leaf rusts. This genotype has high hectolitre weight and sedimentation value with moderate protein content.

PBW 644

Zone: North Western Plains Zone

Production condition: Timely sown, Rainfed/ Restricted irrigated

Parentage	PBW 175/HD 2643
Year of release	2012
Notification No.	1708(E), 26-07-2012
Developed by	PAU, Ludhiana

Morphological features

Plant height (cm)	96
Days to maturity (days)	153
1000-grain weight (g)	40.0
Average yield (q/ha)	31.4
Potential yield (q/ha)	44.8

Quality parameters

Chapati score (max 10)	7.55
Bread loaf volume (cc)	543
Protein content (%)	11.80

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	30.6
Grain yield under N80	32.9



Special Features: PBW 644 has higher yield potential under restricted irrigation conditions and responded well to additional irrigation levels also. This genotype is resistant to stripe & leaf rusts and leaf blight. It has moderate protein content and good chapati score.

North Eastern Plains Zone

Varieties at a glance

Zone	Variety
Timely Sown, Irrigated	NW 5054, K 1006, HD 2967, DBW 39, CBW 38, Raj 4120, K 0307, HD 2824, HD 2733,
Late Sown, Irrigated	DBW 107, HD 3118, HD 2985, HI 1563, NW 2036, DBW 14, HD 2643
Timely Sown, Rainfed	HD 2888, MACS 6145, K 8027, C 306

Package of Practices

Condition	Sowing time	Seed rate	Fertilizer doses and time of application
Timely sown Irrigated	3 rd – 4 th week of November	100 kg/ha	150:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Late sown, Irrigated	2 nd – 3 rd week of December	120 kg/ha	120:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Timely sown Restricted irrigated	4 th week of October - 1 st week of November	100 kg/ha	90:60:40 kg NPK/ha at the time of sowing

NW 5054

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	THELIN//2*ATTILA*2/PASTOR
Year of release	2014
Notification No.	1919 (E), 30-07-2014
Developed by	NDAU&T, Faizabad

Morphological features

Plant height (cm)	102
Days to maturity (days)	122
1000-grain weight (g)	42.1
Average yield (q/ha)	47.0
Potential yield (q/ha)	64.2

Quality feature

Chapati score (max 10)	7.54
Bread loaf volume (cc)	557
Biscuit spread factor (max 10)	7.04

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	44.9
Grain yield under late sown	38.3



Special Features: The variety high yielding ability under optimum planting conditions and has also shown flexibility in planting under late sown conditions up to mid-December. This variety has waxiness, pre-harvest sprouting tolerance and thus suitability to east and far-eastern parts of the country.

K 1006

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	PBW343/HP1731
Year of release	2014
Notification No.	1919 (E), 30-07-2014
Developed by	CSAU&T, Kanpur

Morphological features

Plant height (cm)	86
Days to maturity (days)	123
1000-grain weight (g)	41.1
Average yield (q/ha)	47.0
Potential yield (q/ha)	65.4

Quality parameters

Chapati score (max 10)	7.52
Bread loaf volume (cc)	568
Biscuit spread factor (max 10)	7.72

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	44.0
Grain yield under late sown	35.5



Special Features: K 1006 has erect leaf, non-synchronous tillering and bold grain thus making it suitable for cultivation in eastern India under normal planting situations and has also shown flexibility for late planting. This variety has high protein, good chapati quality and high loaf volume and thus might be suitable for different end products.

HD 2967

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	ALD/COC//URES/HD2160M/HD2278
Year of release	2014
Notification No.	2326(E), 10-10-2011
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	98
Days to maturity (days)	130
1000-grain weight (g)	40.0
Average yield (q/ha)	45.1
Potential yield (q/ha)	65.2

Quality parameters

Chapati score (max 10)	7.99
Bread loaf volume (cc)	548
Biscuit spread factor (max 10)	6.79

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	40.8
Grain yield under late sown	31.8

Special Features: HD 2967 has very wider adaptability for two mega zones (NWPZ & NEPZ). This genotype has shown moderate resistance to stripe and good level of leaf rust resistance. This variety has high zinc, copper and iron thereby securing nutritional security.



DBW 39

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	ATTILA/HUI
Year of release	2010
Notification No.	733(E), 01-04-2010
Developed by	IWBR, Karnal

Morphological features

Plant height (cm)	89
Days to maturity (days)	121
1000-grain weight (g)	42
Average yield (q/ha)	44.6
Potential yield (q/ha)	64.7

Quality parameters

Chapati score (max 10)	7.95
Bread loaf volume (cc)	562
Biscuit spread factor (max 10)	7.27

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	40.1
Grain yield under late sown	31.1

Special Features: DBW 39 has shown very high yield potential under better management conditions. It has shown wider adaptability and stability under timely as well as late sown conditions. This genotype has shown resistance to leaf and stripe rusts and leaf blight. The variety has high chapati score bread quality score, loaf volume and also high biscuit spread factor along-with high zinc and iron.



CBW 38

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	CNDO/R 143/ENTE/MEX1_2/3/Ae.Sq (TAUS)/4/WEAVER/5/2*PASTOR
Year of release	2009
Notification No.	449(E), 11-02-2009
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	93
Days to maturity (days)	121
1000-grain weight (g)	39.1
Average yield (q/ha)	44.4
Potential yield (q/ha)	65.3

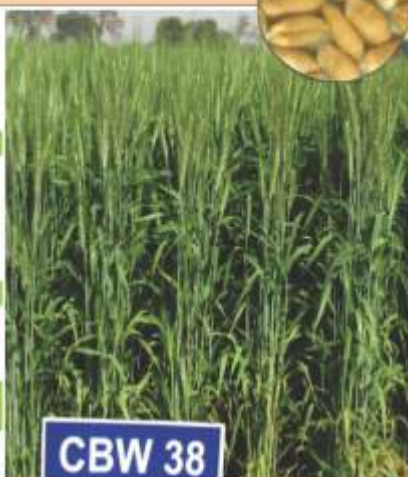
Quality parameters

Chapati score (max 10)	7.64
Bread loaf volume (cc)	570
Biscuit spread factor (max 10)	6.54

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	40.1
Grain yield under late sown	32.3



Special Features: CBW 38 is suitable for varied agronomic conditions and showed minimum reduction ever under late sowing. This variety has potential for heat tolerance in eastern India and also suitable for increased input. This genotype has good chapati and bread making qualities along with high iron and zinc.

RAJ 4120

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	PBW 343/V 1
Year of release	2009
Notification No.	449(E), 11-02-2009
Developed by	RAU, Durgapura

Morphological features

Plant height (cm)	85
Days to maturity (days)	119
1000-grain weight (g)	40.1
Average yield (q/ha)	47.5
Potential yield (q/ha)	66.3

Quality parameters

Chapati score (max 10)	8.01
Bread loaf volume (cc)	562
Biscuit spread factor (max 10)	6.29

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	40.1
Grain yield under late sown	33.0



Special Features: This is medium plant type genotype that has shown good yield potential, disease resistance better adaptability and stability across environments. It showed less reduction under late sown conditions and also resistant against stem rust race Ug99. It has good grain weight and chapati making quality.

K 0307

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	K 8321/UP 2003
Year of release	2007
Notification No.	122(E), 06-02-2007
Developed by	CSAU&T, Kanpur

Morphological features

Plant height (cm)	95
Days to maturity (days)	120
1000-grain weight (g)	39.1
Average yield (q/ha)	45.6
Potential yield (q/ha)	65.1

Quality parameters

Chapati score (max 10)	7.96
Bread loaf volume (cc)	603
Protein content (%)	11.4

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	41.7
Grain yield under late sown	34.3



Special Features: The variety K 0307 has good yielding ability under timely and late sown conditions. This has required level of resistance for leaf rust and leaf blight. The variety has good chapati score and better grain appearance.

HD 2824

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	PTO-1 / CNO 79 / PRL /GAA /3/HD 1951
Year of release	2004
Notification No.	161(E), 04-02-2004
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	89
Days to maturity (days)	124
1000-grain weight (g)	41.0
Average yield (q/ha)	45.7
Potential yield (q/ha)	70.4

Quality parameters

Chapati score (max 10)	6.97
Bread loaf volume (cc)	535
Biscuit spread factor (max 10)	6.69

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	41.7
Grain yield under late sown	32.0



Special Features: HD 2824 has promise for delayed sowing and variable fertilizer doses. It has shown tolerance to leaf rust and leaf blight. This genotype has stiff straw and semi dwarf ideotypes and imparts high degree of tolerance to lodging. The variety has high tillering capacity and erect leaves.

HD 2733

Zone: North Eastern Plains Zone

Production condition: Timely sown, Irrigated

Parentage	PBW226/HW1042//HD 2285
Year of release	2001
Notification No.	92(E), 02-02-2001
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	82
Days to maturity (days)	125
1000-grain weight (g)	41
Average yield (q/ha)	48.1
Potential yield (q/ha)	71.5

Quality parameters

Protein content (%)	11.1
Hectolitre weight (g)	78.1
Sedimentation value	39

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N150	47.2
Grain yield under N180	48.5



Special Features: HD 2733 has high yield potential and wider adaptation (NEPZ and NWPZ) but due to yellow rust susceptibility it is recommended for cultivation in NEPZ only. This genotype has also shown responsiveness to varying irrigation and fertilizers levels thereby indicating for high yield potential under better management conditions. The variety has high tillering capacity and tolerance to lodging.

DBW 107

Zone: North Eastern Plains Zone

Production condition: Late sown, Irrigated

Parentage	TUKURU/INQLAB
Year of release	2015
Notification No.	268 (E), 28-01-2015
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	89
Days to maturity (days)	109
1000-grain weight (g)	39
Average yield (q/ha)	41.3
Potential yield (q/ha)	68.7

Quality parameters

Chapati score (max 10)	7.65
Bread loaf volume (cc)	573
Biscuit spread factor (max 10)	8.06

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	39.65
Grain yield under very late sown	30.86



Special Features: DBW 107 has shown high yield potential under late sown conditions with lower reduction under very late condition thus showing its adaptability to sowing dates. This variety has resistance to leaf rust and blight and thus providing features of tolerance under varying environments. The variety has high protein content, iron and zinc making it rich for nutritional quality.

HD 3118

Zone: North Eastern Plains Zone

Production condition: Late sown, Irrigated

Parentage	ATTILA*2/PBW 65//WBLL1*2/TUKURU
Year of release	2015
Notification No.	268 (E), 28-01-2015
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	94
Days to maturity (days)	112
1000-grain weight (g)	37
Average yield (q/ha)	41.7
Potential yield (q/ha)	66.4

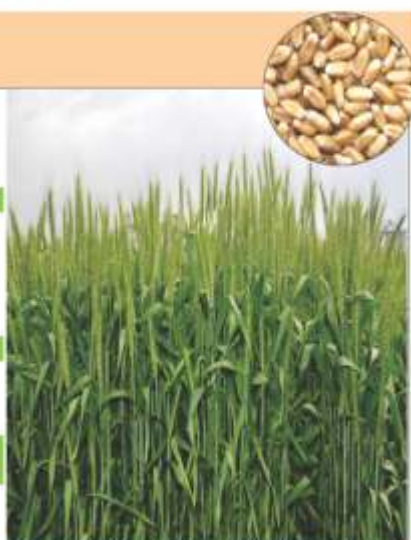
Quality parameters

Chapati score (max 10)	7.52
Bread loaf volume (cc)	568
Biscuit spread factor (max 10)	8.16

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	38.6
Grain yield under very late sown	31.3



Special Features: The variety has shown genetic potential of higher yield under late and very late conditions. This has high levels of resistance against leaf rust and tolerance to leaf blight. The variety has shown higher values of bread loaf volume, protein content and bread quality score.

HD 2985

Zone: North Eastern Plains Zone

Production condition: Late sown, Irrigated

Parentage	PBW 343/PASTOR
Year of release	2011
Notification No.	632(E), 25-03-2011
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	86
Days to maturity (days)	108.3
1000-grain weight (g)	34.1
Average yield (q/ha)	37.7
Potential yield (q/ha)	51.4

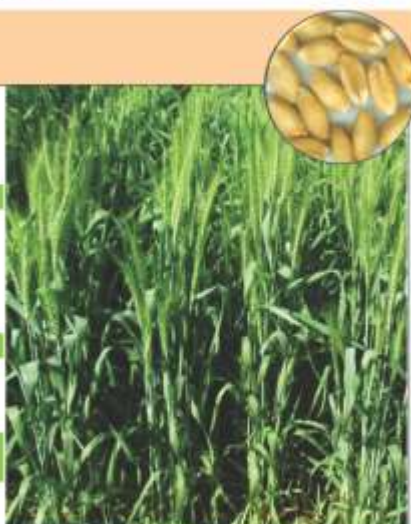
Quality parameters

Chapati score (max 10)	7.63
Bread loaf volume (cc)	565
Biscuit spread factor (max 10)	5.99

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	37.2
Grain yield under very late sown	25.7



Special Features: HD 2985 has shown potential for late and very late planting. It has adult plant resistance against leaf rust and foliar blight. This genotype has more grains per spike alongwith bold grain and is suitable for rice wheat cropping system. This variety has better values of chapati, quality, bread loaf volume and biscuit diameter.

HI 1563

Zone: North Eastern Plains Zone

Production condition: Late sown, Irrigated

Parentage	MACS 2496*2/MC 10
Year of release	2011
Notification No.	622(E), 25-03-2011
Developed by	IARI, Indore

Morphological features

Plant height (cm)	87
Days to maturity (days)	107
1000-grain weight (g)	40.1
Average yield (q/ha)	37.6
Potential yield (q/ha)	51.7

Quality parameters

Chapati score (max 10)	8.01
Bread loaf volume (cc)	555
Biscuit spread factor (max 10)	7.78

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	32.5
Grain yield under very late sown	23.7



Special Features: HI 1563 has wider adaptability, stable high yield, high level of resistance to leaf rust and leaf blight thus making it promising genotype for eastern India. This variety has good grain appearance, chapati, bread and biscuit quality and good levels of micro-nutrients mainly iron, zinc and copper.

NW 2036

Zone: North Eastern Plains Zone

Production condition: Late sown, Irrigated

Parentage	BOW/CROW/BUC/PVN
Year of release	2003
Notification No.	283(E), 12-03-2003
Developed by	NDAU&T, Faizabad

Morphological features

Plant height (cm)	85
Days to maturity (days)	108
1000-grain weight (g)	36.1
Average yield (q/ha)	42.5
Potential yield (q/ha)	66.2

Quality parameters

Protein content (%)	11.53
Hectolitre weight (g)	76.8
Sedimentation value	46.0

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	36.1
Grain yield under very late sown	30.9



Special Features: NW 2036 showed high yield potential for late and very late conditions in eastern India. This genotype also showed tolerance to foliar blight and brown rust diseases. This variety has strong straw and thus tolerant to lodging.

DBW 14

Zone: North Eastern Plains Zone

Production condition: Late sown, Irrigated

Parentage	RAJ 3765/PBW 343
Year of release	2003
Notification No.	283(E), 12-03-2003
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	89
Days to maturity (days)	102
1000-grain weight (g)	40
Average yield (q/ha)	42.7
Potential yield (q/ha)	63.9

Quality parameters

Chapati score (max 10)	6.75
Bread loaf volume (cc)	580
Bread quality (max 10)	7.56

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	35.1
Grain yield under very late sown	30.2

Special Features: DBW 14 has shown high yield for late sown condition in rice wheat cropping system. The genotype has shown minimum reduction under very late sowing and hence was heat tolerant. This variety was tolerant to yellow and brown rust and also to leaf blight. Also it has high protein, good grain appearance, good Chapati and bread making quality.



HD 2643

Zone: North Eastern Plains Zone

Production condition: Irrigated, late sown

Parentage	VEE'S/HD 2407 //HD 2329
Year of release	1997
Notification No.	360(E)
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	82
Days to maturity (days)	111
1000-grain weight (g)	41.2
Average yield (q/ha)	36.4
Potential yield (q/ha)	53.9

Quality parameters

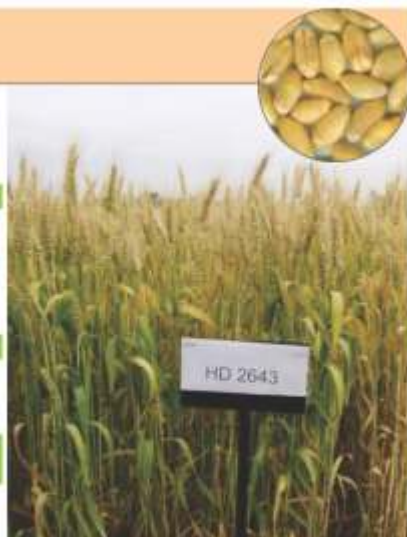
Chapati score (max 10)	7.7
Protein content (%)	13.1
Sedimentation value (ml)	45.5

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	33.8
Grain yield under late sown	33.4

Special Features: HD 2643 has shown high yield under late planting in eastern India during the years. This variety showed resistance against all the three rusts and Karnal bunt. This variety has bold grain and good chapati making quality.



HD 2888

Zone: North Eastern Plains Zone

Production condition: Timely sown, Rainfed

Parentage	C 306/T.sphaerococcum//HW 2004
Year of release	2006
Notification No.	599(E), 25-04-2006
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	97
Days to maturity (days)	125
1000-grain weight (g)	39
Average yield (q/ha)	22.5
Potential yield (q/ha)	38.3

Quality parameters

Chapati score (max 10)	7.71
Bread loaf volume (cc)	525
Biscuit spread factor (max 10)	6.10

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	21.7
Grain yield under N80	22.7



Special Features: HD 2888 has shown yield superiority under rainfed conditions of eastern India. IT has also shown high degree of adult plant resistance to leaf rust and moderate tolerance to foliar blight. The genotype.

MACS 6145

Zone: North Eastern Plains Zone

Production condition: Timely sown, Rainfed

Parentage	C 306+ Lr28 (HW 2034)
Year of release	2002
Notification No.	122(E), 02-02-2005
Developed by	ARI, Pune

Morphological features

Plant height (cm)	118
Days to maturity (days)	127
1000-grain weight (g)	40
Average yield (q/ha)	25.5
Potential yield (q/ha)	37.0

Quality parameters

Chapati score (max 10)	6.3
Protein Content (%)	11.4
Bread Quality Score (max 10)	4.74

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	23.1
Grain yield under N80	25.4



Special Features: MACS 6145 is isogenic line of C 306 alongwith Lr28, possesses resistance to leaf rust and Quality parameters for better chapati and bread score were recorded. This variety has shown good potential under varying levels of fertilizers has thermo-tolerance and lustrous grains,

K 8027

Zone: North Eastern Plains Zone

Production condition: Timely sown, Rainfed

Parentage	NP875/4/N10B/Y53//Y50/3/KT54B/5/ 2*K852
Year of release	1989
Notification No.	599(E), 31-07-1989
Developed by	CSAU&T, Kanpur

Morphological features

Plant height (cm)	104
Days to maturity (days)	140
1000-grain weight (g)	43.0
Average yield (q/ha)	25.1
Potential yield (q/ha)	37.1

Quality parameters

Chapati score (max 10)	8.04
Bread loaf volume (cc)	490
Biscuit spread factor (max 10)	7.73

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	-
Grain yield under N80	-

Special Features: K8027 is a potential genotype for timely sown rainfed conditions of NEPZ. It is medium tall, bold grain genotype suitable for chapati making.



C 306

Zone: North Eastern Plains Zone

Production condition: Timely sown, Rainfed

Parentage	RGN/CSK3 //2*C591/3/C217/N14 //C281
Year of release	1965
Notification No.	4045, 24-09-1969
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	111
Days to maturity (days)	137
1000-grain weight (g)	41.0
Average yield (q/ha)	26.0
Potential yield (q/ha)	36.0

Quality parameters

Chapati score (max 10)	8.10
Bread loaf volume (cc)	496
Biscuit spread factor (max 10)	7.79

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	-
Grain yield under N80	-

Special Features: C 306 is bread wheat genotype for timely sown rainfed conditions of northern India (NWPZ and NEPZ). It is a medium tall light green and semi-erect genotype having high protein and high harvest index. This variety is most preferred and suitable for chapati making.



Central Zone

Varieties at a glance

Condition	Variety
Timely sown, Irrigated	Bread Wheat : WH 1142, HI 1544, GW 366, GW 322, GW 273 Durum Wheat : HI 8737, HI 8713, MPO 1215, HI 8498
Late sown, Irrigated	MP 3336, MP 1203, HD 2932, HD 2864, MP 4010
Timely sown, Rainfed/ Restricted irrigated	Bread Wheat : DBW 110, MP 3288, MP 3173, HI 1531, HI 1500 Durum Wheat : HD 4672

Package of Practices

Condition	Sowing time	Seed rate	Fertilizer doses and time of application
Timely sown, Irrigated	2 nd – 3 rd week of November	100 kg/ha	120:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Late sown, Irrigated	1 st - 2 nd week of December	120 kg/ha	90:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Timely sown, Restricted irrigated	4 th week of October - 1 st week of November	120 kg/ha	90:60:40 kg NPK/ha at the time of sowing
Timely sown, Rainfed	3 rd – 4 th week of October	100 kg/ha	60:30:20 kg NPK/ha at the time of sowing

WH 1142

Zone: Central Zone

Production condition: Timely sown, Irrigated

Parentage	CHEN/Ae.Sq. (TAUS) //FCT/3/2*WEAVER
Year of release	2014
Notification No.	1228(E), 07-05-2015
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	102
Days to maturity (days)	154
1000-grain weight (g)	38
Average yield (q/ha)	48.1
Potential yield (q/ha)	62.5

Quality parameters

Chapati score (max 10)	7.49
Bread loaf volume (cc)	568
Biscuit spread factor (max 10)	7.50

Agronomic practices

Grain yield (q/ha) under different irrigation levels

Grain yield under one irrigation	44.9
Grain yield under two irrigations	47.5

Special Features: WH 1142 has shown yield potential under restricted irrigation conditions of North Western Plains Zone. This genotype was found tolerant to yellow and brown rust and lodging as well. Results indicated that it has good quality parameters namely gluten (%), bread loaf vol., and yellow pigment.



HI 1544

Zone: Central Zone

Production condition: Timely sown, Irrigated

Parentage	HINDI 62/BOBWHITE/CPAN 2099
Year of release	2008
Notification No.	1108(E), 08-05-2008
Developed by	IARI, Indore

Morphological features

Plant height (cm)	87
Days to maturity (days)	111
1000-grain weight (g)	42
Average yield (q/ha)	51.4
Potential yield (q/ha)	68.2

Quality parameters

Chapati score (max 10)	8.10
Bread loaf volume (cc)	522
Biscuit spread factor (max 10)	6.06

Agronomic practices

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	47.3
Grain yield under late sown	35.5

Special Features: HI 1544 combines early maturity, high stable yield, tolerance to terminal stress along with resistance to stem and leaf rust. It has high hectolitre weight and sedimentation value for good chapati making quality.



GW 366

Zone: Central Zone

Production condition: Timely sown, Irrigated

Parentage	DL 802-3/GW 232
Year of release	2007
Notification No.	122(E), 06-02-2007
Developed by	IAU, Junagarh

Morphological features

Plant height (cm)	92
Days to maturity (days)	117
1000-grain weight (g)	49.0
Average yield (q/ha)	51.7
Potential yield (q/ha)	77.9

Quality parameters

Chapati score (max 10)	7.78
Bread loaf volume (cc)	552
Biscuit spread factor (max 10)	6.64

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	48.7
Grain yield under late sown	42.5



Special Features: GW 366 is a high yielding variety and has shown adaptability to different dates of sowing. This variety is resistant to leaf and stem rusts. It has good grain quality parameters, bold grain and high protein content.

GW 322

Zone: Central zone

Production condition: Timely sown, Irrigated

Parentage	GW 173/GW 196
Year of release	2002
Notification No.	937(E), 04-09-2002
Developed by	SAAU, Junagarh

Morphological features

Plant height (cm)	84
Days to maturity (days)	112
1000-grain weight (g)	40.3
Average yield (q/ha)	46.9
Potential yield (q/ha)	66.3

Quality parameters

Protein content (%)	11.3
Hectolitre weight (kg)	79.4
Sedimentation value (ml)	38.0

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	45.1
Grain yield under late sown	39.2



Special Features: GW 322 has shown high yield potential for timely and late planting in central zone. It is a nutrient dense genotype and has drought tolerance with increased water use efficiency and therefore suitability for limited irrigation conditions also. This variety has excellent grain appearance and good protein content.

GW 273

Zone: Central zone

Production condition: Timely sown, Irrigated

Parentage	CPAN 2084/VW 205
Year of release	1998
Notification No.	401(E), 15-05-1998
Developed by	IAU, Vijapur

Morphological features

Plant height (cm)	96
Days to maturity (days)	113
1000-grain weight (g)	47
Average yield (q/ha)	44.7
Potential yield (q/ha)	64.8

Quality parameters

Protein content (%)	11.7
Hectolitre weight (g)	80.4
Sedimentation value	62

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	52.1
Grain yield under late sown	47.8



Special Features: GW 273 has shown yielding ability under better management condition in central India. It has excellent grain quality and required level of leaf resistance. Being an early maturing genotype, it escapes from high temperature stress.

HI 8737 (Durum)

Zone: Central zone

Production condition: Timely sown, Irrigated

Parentage	HI 8177/ HI 8158 // HI 8498
Year of release	2014
Notification No.	268 (E), 28-01-2015
Developed by	IARI, INDORE

Morphological features

Plant height (cm)	85
Days to maturity (days)	124
1000-grain weight (g)	51
Average yield (q/ha)	53.4
Potential yield (q/ha)	81.0

Quality parameters

Yellow pigment (ppm)	5.38
Grain hardness Index (%)	70.1
Protein Content (%)	12.1

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	51.1
Grain yield under late sown	41.7



Special Features: HI 8737 is a widely adopted high yielding durum genotype that has shown yield stability under timely and late sown conditions. The variety has good level of resistance against leaf and stem rust and seedling resistance test indicated presence of diverse genes for resistance. The variety is good for chapati making and also for pasta preparations and essential micro-nutrients like iron and zinc.

HI 8713 (Durum)

Zone: Central zone

Production condition: Timely sown, Irrigated

Parentage	HD 4672 / PDW 233
Year of release	2013
Notification No.	952(E), 10-04-2013
Developed by	IARI, Indore

Morphological features

Plant height (cm)	93
Days to maturity (days)	122
1000-grain weight (g)	45
Average yield (q/ha)	52.3
Potential yield (q/ha)	68.2

Quality parameters

Yellow pigment (ppm)	7.16
Grain Hardness Index (%)	70.0
Protein content (%)	11.7

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	52.5
Grain yield under late sown	42.8



Special Features: This is a widely adopted durum wheat genotype suitable for timely and late sown conditions. It has acceptable level of resistance but very good quality parameters for end products as well as micronutrients like iron, zinc, copper and manganese.

MPO 1215 (Durum)

Zone: Central Zone

Production condition: Timely sown, Irrigated

Parentage	GW 1113/ GW 1114//HI 8381
Year of release	2010
Notification No.	733(E), 01-04-2010
Developed by	JNKVV, Powarkheda

Morphological features

Plant height (cm)	82
Days to maturity (days)	115
1000-grain weight (g)	51
Average yield (q/ha)	45.1
Potential yield (q/ha)	65.3

Quality parameters

Yellow pigment (ppm)	4.68
Protein content (%)	12.1
Hectolitre weight (g)	83.8

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	45.5
Grain yield under late sown	40.1



Special Features: Durum wheat variety MPO 1215 is an early maturing genotype with adaptability to varying dates of sowing. This variety has very high test weight, rich in zinc, iron and copper and good pasta making quality.

HI 8498 (Durum)

Zone: Central Zone

Production condition: Timely sown, Irrigated

Parentage	CR 'S'-GS'S' /A-9-30-1/RAJ 911
Year of release	1999
Notification No.	425(E), 08-06-1999
Developed by	IARI, Indore

Morphological features

Plant height (cm)	98
Days to maturity (days)	120
1000-grain weight (g)	50.0
Average yield (q/ha)	44.3
Potential yield (q/ha)	73.0

Quality parameters

Protein content (%)	12.4
Yellow pigment (ppm)	4.3
Hectolitre weight (g)	82.1

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	52.5
Grain yield under late sown	44.4

Special Features: HI 8498 is a high yielding, stable and fertilizer responsive genotype possessing resistance to rust (Stem and brown). This variety has good grain quality, pasta making attributes and thus a durum genotype that fetched better price in the market.



MP 3336

Zone: Central Zone

Production condition: Late sown, Irrigated

Parentage	HD 2402/GW 173
Year of release	2013
Notification No.	2817 (E), 19-09-2013
Developed by	JABALPUR

Morphological features

Plant height (cm)	81
Days to maturity (days)	107
1000-grain weight (g)	42.0
Average yield (q/ha)	44.7
Potential yield (q/ha)	64.4

Quality parameters

Chapati score (max 10)	8.02
Bread loaf volume (cc)	535
Biscuit spread factor (max 10)	6.91

Agronomic practices-

Grain yield (q/ha) under late / very late planting

Grain yield under late sown	45.1
Grain yield under very late sown	32.6

Special Features: Wheat variety MP 3336 has shown high yield potential and stability over environments (Late and very late conditions). This variety is early in heading and maturity and thus has chances to do better under limited irrigation conditions. It is tolerant to black and leaf rust, has good grain appearance, better bread making quality and good levels of micronutrients.



MP 1203

Zone: Central Zone

Production condition: Late sown, Irrigated

Parentage	FASN/28TEPOKA/3/CHEN/ Ae.Sq./TA
Year of release	2009
Notification No.	454(E), 11-02-2009
Developed by	JNKVV, Powarkheda

Morphological features

Plant height (cm)	85
Days to maturity (days)	110
1000-grain weight (g)	39
Average yield (q/ha)	41.2
Potential yield (q/ha)	59.6

Quality parameters

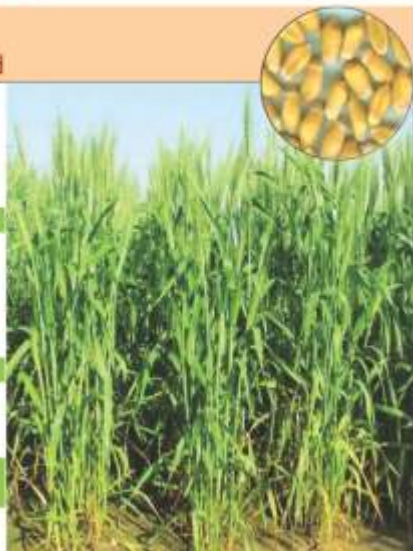
Chapati score (max 10)	7.55
Bread loaf volume (cc)	580
Biscuit spread factor (max 10)	6.45

Agronomic practices-

Grain yield (q/ha) under late / very late planting

Grain yield under late sown	43.0
Grain yield under very late sown	33.0

Special Features: This variety has shown good potential for late and very late conditions of central zone. This also has resistance to diseases and very high protein content with high values of micronutrients.



HD 2932

Zone: Central Zone

Production condition: Late sown, Irrigated

Parentage	KAUZ/STAR//HD 2643
Year of release	2008
Notification No.	1108(E), 08-05-2008
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	84
Days to maturity (days)	109
1000-grain weight (g)	43
Average yield (q/ha)	42.2
Potential yield (q/ha)	57.8

Quality parameters

Chapati score (max 10)	7.96
Bread loaf volume (cc)	541
Biscuit spread factor (max 10)	6.29

Agronomic practices-

Grain yield (q/ha) under late / very late planting

Grain yield under late sown	41.2
Grain yield under very late sown	31.8

Special Features: HD 2932 has shown wider adaptability across the zones with high yield potential. It has APR against brown and black rusts. This genotype has more number of grains and high thousand grain weight even under late sowings, thus indicating to soil temperature stress. It also high iron and zinc and good chapati making quality.



HD 2864

Zone: Central Zone

Production condition: Late sown, Irrigated

Parentage	DL 509-2/ DL 377-8
Year of release	2005
Notification No.	122(E), 02-02-2005
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	81
Days to maturity (days)	107
1000-grain weight (g)	38.1
Average yield (q/ha)	41.7
Potential yield (q/ha)	51.4

Quality parameters

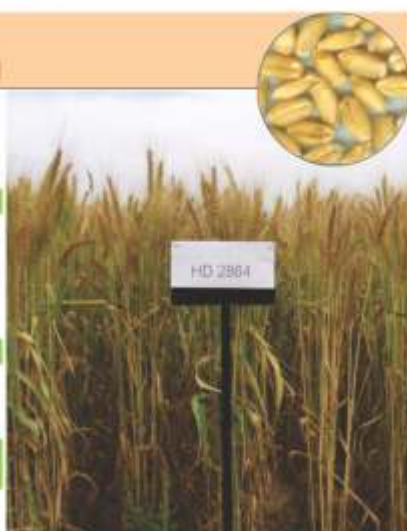
Chapati score (max 10)	7.54
Bread loaf volume (cc)	550
Bread quality (max 10)	6.71

Agronomic practices-

Grain yield (q/ha) under late / very late planting

Grain yield under late sown	39.4
Grain yield under very late sown	25.5

Special Features: HD 2864 has yield stability over years and locations and high yield under very late conditions. This has high protein good grain appearance and good combination for chapati bread making quality. This genotype has more flour recovery and hence better storage ability.



MP 4010

Zone: Central Zone

Production condition: Late sown, Irrigated

Parentage	ANGOSTURA 88
Year of release	2003
Notification No.	283(E), 12-03-2003
Developed by	JNKVV, Gwalior

Morphological features

Plant height (cm)	75
Days to maturity (days)	108
1000-grain weight (g)	43
Average yield (q/ha)	40.1
Potential yield (q/ha)	53.6

Quality parameters

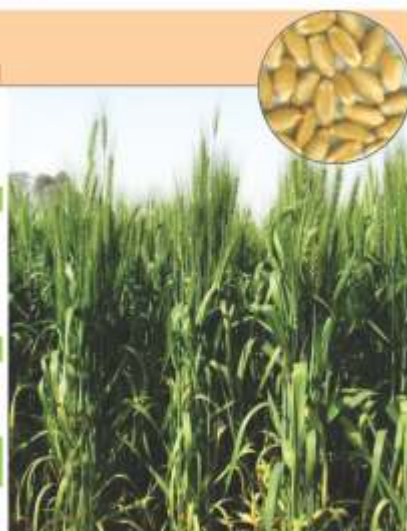
Protein content (%)	12.6
Hectolitre weight (g)	82.1
Sedimentation value	48.5

Agronomic practices-

Grain yield (q/ha) under late / very late planting

Grain yield under late sown	38.8
Grain yield under very late sown	24.4

Special Features: MP 4010 has high yield potential for late and very late conditions and also possesses tolerance to high temperature. This genotype possesses resistance to both leaf and stem rust, has bold seed and excellent grain.



DBW 110

Zone: Central Zone

Production condition: Timely sown, Restricted irrigated

Parentage	KIRITATI/4/25ER118*2/3KAUZ*2/8OW//KAUZ
Year of release	2015
Notification No.	268 (E), 28-01-2015
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	89
Days to maturity (days)	124
1000-grain weight (g)	43
Average yield (q/ha)	39.2
Potential yield (q/ha)	50.1

Quality parameters

Chapati score (max 10)	7.83
Bread loaf volume (cc)	555
Biscuit spread factor (max 10)	6.59

Agronomic practices-

Grain yield (q/ha) under different irrigations levels

Grain yield under one irrigation	32.0
Grain yield under two irrigation	36.2



Special Features: DBW 110 has high yield potential under timely sown restricted irrigation condition in peninsular India. This genotype is resistant to brown and black rust. It is also resistant to Karnal bunt (KB). This genotype has shown promise for better yield with two irrigations. It has good protein, test weight and good chapati/ bread making score.

MP 3288

Zone: Central Zone

Production condition: Timely sown, Rainfed/Restricted irrigation

Parentage	DOVE/BUC/DL 7882
Year of release	2011
Notification No.	2326(E), 10-10-2011
Developed by	JNKVV, Jabalpur

Morphological features

Plant height (cm)	75.0(RF), 89.3 (RI)
Days to maturity (days)	121 (RF), 123 (RI)
1000-grain weight (g)	41.0 (RF), 44.0 (RI)
Average yield (q/ha)	23.2 (RF), 35.1(RI)
Potential yield (q/ha)	42.1 (RF), 43.9 (RI)

Quality parameters

Chapati score (max 10)	7.87(RF), 7.95(RI)
Bread loaf volume (cc)	562(RF), 560(RI)
Biscuit spread factor (max 10)	6.45(RF), 7.40(RI)

Agronomic practices-

Grain yield (q/ha) under different irrigations levels

Grain yield under one irrigation	32.6
Grain yield under two irrigation	37.7



Special Features: MP 3288 is an early maturing, semi dwarf, non-lodging genotype suitable for rainfed as well as restricted irrigation conditions. This has resistance to leaf and black rusts. This variety has high bread loaf volume and good chapati making attributes.

MP 3173

Zone: Central Zone

Production condition: Timely sown, Restricted irrigated

Parentage	HI 1011/WH 965-1
Year of release	2009
Notification No.	449(E), 11-02-2009
Developed by	JNKVV, Jabalpur

Morphological features

Plant height (cm)	91
Days to maturity (days)	128
1000-grain weight (g)	43.1
Average yield (q/ha)	25.7
Potential yield (q/ha)	46.0

Quality parameters

Protein content (%)	11.2
Hectolitre weight (g)	81.3
Grain appearance	6.1

Agronomic practices-

Grain yield (q/ha) under different irrigations levels

Grain yield under one irrigation	20.2
Grain yield under two irrigation	25.5

Special Features: MP 3173 is a semi dwarf genotype with thick culm and hard threshability thus tolerant to lodging and grain shattering. This variety shown higher degree of resistant to rusts and other diseases of India and has shown stable performance for both rainfed and limited irrigation conditions.



HI 1531

Zone: Central Zone

Production condition: Timely sown, Restricted irrigated

Parentage	HI 1182/CPAN 1990
Year of release	2005
Notification No.	599(E), 25-04-2006
Developed by	IARI, Indore

Morphological features

Plant height (cm)	72 (RF), 93(RI),
Days to maturity (days)	123(RF), 131(RI)
1000-grain weight (g)	40 (RF), 41(RI)
Average yield (q/ha)	23.6(RF), 26.6(RI)
Potential yield (q/ha)	39.8(RF), 40.0(RI)

Quality parameters

Chapati score (max 10)	7.61
Bread loaf volume (cc)	287
Biscuit spread factor (max 10)	7.44

Agronomic practices-

Grain yield (q/ha) under different irrigations levels

Grain yield under one irrigation	21.2
Grain yield under two irrigation	27.1

Special Features: HI 1531 is early maturing semi dwarf, lodging resistant and drought tolerant bread wheat genotype for central India. This also has high resistance to leaf and stem rusts thereby ensuring high yield with better protection.



HI 1500

Zone: Central Zone

Production condition: Timely sown, Rainfed

Parentage	HW 2002*2//STREMPALLI/PNC 5
Year of release	2003
Notification No.	283(E), 12-03-2003
Developed by	IARI, Indore

Morphological features

Plant height (cm)	83
Days to maturity (days)	124
1000-grain weight (g)	42.0
Average yield (q/ha)	16.1
Potential yield (q/ha)	30.3

Quality parameters

Protein content (%)	11.6
Hectolitre weight (g)	88.0
Grain appearance	6.9

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	16.3
Grain yield under N80	17.2



Special Features: HI 1500 is the early maturing bread wheat genotype possessing drought tolerance for better yield under limited input conditions. This genotype has shown resistance to leaf and stem rusts. It also has high hectolitre weight and good chapati Quality parameters.

HD 4672 (Durum)

Zone: Central Zone

Production condition: Timely sown, Rainfed

Parentage	BIJAGA RED /PBW34 //ALTAR 84
Year of release	2000
Notification No.	340(E), 03-04-2000
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	68
Days to maturity (days)	125
1000-grain weight (g)	45
Average yield (q/ha)	15.1
Potential yield (q/ha)	34.2

Quality parameters

Yellow pigment (ppm)	4.9
Protein content (%)	11.6
Hectolitre weight (g)	81

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N90	14.5
Grain yield under N120	13.5



Special Features: HD 4672 is a semi-dwarf disease resistant and good quality durum genotype for timely sown rainfed low fertility conditions of central zone. This genotype has shown promise for limited irrigation.

Peninsular Zone

Varieties at a glance

Condition	Variety
Timely sown, Irrigated	Bread Wheat : MACS 6478, UAS 304, MACS 6222, NIAW 917, Raj 4037, GW 322 Durum Wheat : UAS 428, WHD 948, UAS 415 Dicoccum Wheat : MACS 2971, DDK 1029 DDK 1025
Late sown, Irrigated	HD 3090, AKAW 4627, HD 2932, Raj 4083, HD 2833
Timely sown, Rainfed/ Restricted irrigated	DBW 93, NIAW 1415, HD 2987, HD 2781, UAS 446

Package of Practices

Condition	Sowing time	Seed rate	Fertilizer doses and time of application
Timely sown, Irrigated	2 nd – 3 rd week of November	100 kg/ha	120:60:40 kg NPK /ha (1/3N at sowing and remaining in two equal splits at first and second irrigation)
Late sown, Irrigated	1 st - 2 nd week of December	120 kg/ha	90:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)
Timely sown, Restricted irrigated	1 st week of November	100 kg/ha	90:60:40 kg NPK/ha at the time of sowing
Timely sown, Rainfed	3 rd – 4 th week of October	100 kg/ha	60:30:20 kg NPK/ha at the time of sowing

MACS 6478

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	CS/TH.SC//3*PVN/3MIRLO/BUC/4/ MILAN/5/TILHI
Year of release	2014
Notification No.	1919 (E), 30-07-2014
Developed by	ARI, Pune

Morphological features

Plant height (cm)	78
Days to maturity (days)	105
1000-grain weight (g)	45.1
Average yield (q/ha)	45
Potential yield (q/ha)	65.7

Quality parameters

Chapati score (max 10)	8.05
Bread loaf volume (cc)	568
Biscuit spread factor (max 10)	6.22

Agronomic practices

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	37.4
Grain yield under late sown	30.7



Special Features: Variety MACS 6478 is highly resistant to both black and brown rust and carries adult plant resistance genes. This variety has high yield potential and thus can give better yield with matching agronomic practices. The variety has high protein content and better nutritional quality for zinc and iron thereby making it good for chapati and bread quality.

UAS 304

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	SERI/CEP80120//KAUZ/PBW 343
Year of release	2013
Notification No.	2817 (E), 19-09-2013
Developed by	UAS, Dharwad

Morphological features

Plant height (cm)	80
Days to maturity (days)	109
1000-grain weight (g)	41.7
Average yield (q/ha)	46.8
Potential yield (q/ha)	59.9

Quality parameters

Protein content	12.4
Sedimentation value	41.3
Hectolitre weight	81

Agronomic practices

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	42.90
Grain yield under late sown	29.14



Special Features: UAS 304 is widely adopted genotype of bread wheat possessing multiple disease resistance to prevailing leaf and stem rust races and spot blotch. This variety good protein, bold grain and acceptable grain quality.

MACS 6222

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	HD 2189*2//MACS 2496
Year of release	2010
Notification No.	733(E), 01-04-2010
Developed by	ARI, Pune

Morphological features

Plant height (cm)	82
Days to maturity (days)	106
1000-grain weight (g)	44.3
Average yield (q/ha)	47.7
Potential yield (q/ha)	60.9

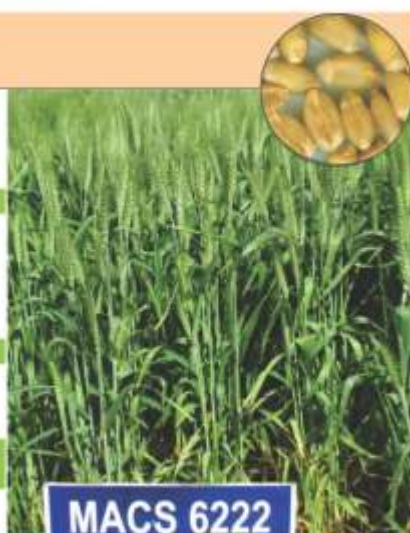
Quality parameters

Chapati score (max 10)	7.45
Bread loaf volume (cc)	605
Biscuit spread factor (max 10)	6.49

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	45.3
Grain yield under late sown	41.2



Special Features: MACS 6222 is a high yielding genotype with early maturity and resistance to prevalent leaf and stem rust races. This variety also has adaptability to late sowings having least reduction thus suitable for varying dates of sowing. It has bold grain, good Quality parameters and rich in micro nutrients availability.

NIAW 917

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	GW 244/BOB WHITE
Year of release	2006
Notification No.	599(E), 25-04-2006
Developed by	ARS, Niphad

Morphological features

Plant height (cm)	86
Days to maturity (days)	105
1000-grain weight (g)	38.0
Average yield (q/ha)	46.6
Potential yield (q/ha)	62.5

Quality parameters

Chapati score (max 10)	7.48
Bread loaf volume (cc)	586
Biscuit spread factor (max 10)	6.57

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	44.8
Grain yield under late sown	43.6



Special Features: This variety has shown higher yielding ability under normal and late sown conditions and thus wide adaptation. It has also shown resistance to leaf rust. This variety is good for chapati & bread quality and essential micro-nutrient including iron and zinc.

RAJ 4037

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	DL 788-2/RAJ 3717
Year of release	2004
Notification No.	161(E), 04-02-2004
Developed by	RARS, Durgapura

Morphological features

Plant height (cm)	74
Days to maturity (days)	104
1000-grain weight (g)	44
Average yield (q/ha)	40.9
Potential yield (q/ha)	59.3

Quality parameters

Protein content (%)	11.8
Sedimentation value (ml)	43.3
Hectolitre weight (g)	82.5

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	38.3
Grain yield under late sown	35.0

Special Features: Variety Raj 4037 has potential for higher yield following recommended agronomy and showed minimum reduction under late sown. This variety possesses resistance to black and brown rusts and has excellent grain. It has inbuilt mechanism to yield higher under warmer areas and matures early in heat stress environment.



GW 322

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	GW 173/GW 196
Year of release	2002
Notification No.	937(E), 04-09-2002
Developed by	SDAU, Vijapur

Morphological features

Plant height (cm)	83
Days to maturity (days)	110
1000-grain weight (g)	43.0
Average yield (q/ha)	41.7
Potential yield (q/ha)	61.8

Quality parameters

Protein content (%)	11.5
Hectolitre Weight (g)	80.1
Sedimentation value (ml)	35.4

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	42.3
Grain yield under late sown	35.3

Special Features: GW 322 showed high yield in central zone for irrigated timely conditions. The genotype has also done well under late sowing and indicated tolerance to terminal heat stress. This variety has capacity to make better use of nutrients and also showed tolerance to drought stress. This genotype had resistance to both black and leaf rusts with diverse gene.



GW 322

UAS 428 (Durum)

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	GREEN-14/YAV-10/AUK/UAS 402
Year of release	2012
Notification No.	1708(E), 26-07-2012
Developed by	UAS, Dharwad

Morphological features

Plant height (cm)	83
Days to maturity (days)	108
1000-grain weight (g)	43.7
Average yield (q/ha)	48.0
Potential yield (q/ha)	58.8

Quality parameters

Yellow pigment (ppm)	5.49
Grain hardness index (%)	80.0
Protein content (g)	12.4

Agronomic practices

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	42.5
Grain yield under late sown	37.3

Special Features: Durum wheat genotype UAS 428 has high yielding potential and better adaptability to sowing periods. This variety possesses multiple disease resistance to prevailing rust races (Brown and black) and spot blotch. The variety has bold grain and good end product alongwith essential micro elements.



WHD 948 (Durum)

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	Altar 84/Stint//Silver
Year of release	2012
Notification No.	1146 (E), 24-04-2014
Developed by	CCSHAU, Hisar

Morphological features

Plant height (cm)	83
Days to maturity (days)	109
1000-grain weight (g)	49.1
Average yield (q/ha)	46.5
Potential yield (q/ha)	69.5

Quality parameters

Yellow pigment (ppm)	5.99
Sedimentation value (ml)	30.1
Protein content (%)	12.7

Agronomic practices

Grain yield (q/ha) under normal / planting

Grain yield under normal sown	43.6
Grain yield under late sown	40.4

Special Features: WHD 948 is a widely adopted durum wheat genotype suitable for irrigated, timely sown conditions of Peninsular zone (Maharashtra and Karnataka) of India. This variety showed better resistance against leaf and stem rusts. The variety has high yellow pigment, better nutritional elements like manganese and suitable for pasta products.



UAS 415 (Durum)

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	GALLI/BOOMER20//WR 185
Year of release	2008
Notification No.	454(E), 11-02-2009
Developed by	UAS, Dharwad

Morphological features

Plant height (cm)	84
Days to maturity (days)	112
1000-grain weight (g)	43.7
Average yield (q/ha)	49.6
Potential yield (q/ha)	61.8

Quality parameters

Protein content (%)	11.4
Yellow pigment (ppm)	4.49
Hectolitre weight (g)	82

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	51.0
Grain yield under late sown	39.1

Special Features: UAS 415 has shown high yield for timely and late sowing in peninsular India and thus wider adaptation. It has multiple disease resistance for leaf and stem rusts, foliar blight, powdery mildew and flag smut.



MACS 2971 (Dicoccum)

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	KTR 5*2/NP 200
Year of release	2009
Notification No.	2187(E), 27-08-2009
Developed by	ARI, Pune

Morphological features

Plant height (cm)	89
Days to maturity (days)	107
1000-grain weight (g)	41
Average yield (q/ha)	50.2
Potential yield (q/ha)	62.2

Quality parameters

Protein Content (%)	12.9
Yellow pigment (ppm)	3.23
Sedimentation value	20.0

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	44.8
Grain yield under late sown	39.4

Special Features: Dicoccum variety MACS 2971 showed high yield under timely and late planting. It has high resistance to stem and leaf rusts. This genotype has high protein content, good grain appearance and overall end product quality.



DDK 1029 (Dicoccum)

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	DDK 1012/HW 1093//276-15
Year of release	2007
Notification No.	1703(E), 05-10-2007
Developed by	UAS, Dharwad

Morphological features

Plant height (cm)	81
Days to maturity (days)	103
1000-grain weight (g)	43.0
Average yield (q/ha)	45.6
Potential yield (q/ha)	59.9

Quality parameters

Yellow pigment (ppm)	3.9
Protein content (%)	13.0
Sedimentation value	26.0

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	47.4
Grain yield under late sown	42.9



Special Features: DDK 1029 has high yield and wider adaptability in central, peninsular and southern hill zones. It has adaptability to both timely and late sown conditions and high effective tillers. Genotype is good for yellow pigment and grain score.

DDK 1025 (Dicoccum)

Zone: Peninsular Zone

Production condition: Timely sown, Irrigated

Parentage	DDK 1013/DDK 1001//278-13
Year of release	2006
Notification No.	599(E), 25-04-2006
Developed by	UAS, Dharwad

Morphological features

Plant height (cm)	89
Days to maturity (days)	107
1000-grain weight (g)	41.8
Average yield (q/ha)	38.1
Potential yield (q/ha)	49.7

Quality parameters

Protein content (%)	13.0
Yellow pigment (ppm)	3.8
Sedimentation value	33.2

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	41.7
Grain yield under late sown	38.8



Special Features: DDK 1025 has high yield potential and wider adaptability across zones. It has also shown least reduction under late planting and was found suitable for increased level of fertilizers and irrigations. This variety is good for nutritional quality, dietary fiber and thus good for carbohydrate digestibility.

HD 3090

Zone: Peninsular Zone

Production condition: Late sown, Irrigated

Parentage	SFW/VAISHALI//UP 2425
Year of release	2014
Notification No.	244 (E), 24-01-2014
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	80
Days to maturity (days)	101
1000-grain weight (g)	37.2
Average yield (q/ha)	41.4
Potential yield (q/ha)	63.1

Quality parameters

Chapati score (max 10)	7.55
Bread loaf volume (cc)	575
Biscuit spread factor (max 10)	7.73

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	34.8
Grain yield under very late sown	29.9



Special Features: This genotype is an early maturing semi dwarf with high yield potential under irrigated late sown conditions. It has high degree of resistance against leaf and stem rusts. The variety has high protein content, high biscuit spread factor and also good for chapati making.

AKAW 4627

Zone: Peninsular Zone

Production condition: Late sown, Irrigated

Parentage	WH 147/SUNSTAR*/C 80.1(SELECTION FROM VIMAL)
Year of release	2010
Notification No.	456(E), 16-03-2012
Developed by	PDKV, Akola

Morphological features

Plant height (cm)	78
Days to maturity (days)	96
1000-grain weight (g)	40
Average yield (q/ha)	42.8
Potential yield (q/ha)	60.9

Quality parameters

Chapati score (max 10)	7.53
Protein Content (%)	12.3
Sedimentation value	45

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	40.9
Grain yield under very late sown	35.2



Special Features: This is a bread wheat genotype for irrigated late sown conditions of peninsular zone and has high yield potential even under late sowings. This variety exhibited resistance against black and brown rusts races. It has better quality for chapati, bread and biscuit.

HD 2932

Zone: Peninsular Zone

Production condition: Late sown, Irrigated

Parentage	KAUZ/STAR//HD 2643
Year of release	2008
Notification No.	1108(E), 08-05-2008
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	85
Days to maturity (days)	99
1000-grain weight (g)	43
Average yield (q/ha)	43.3
Potential yield (q/ha)	53.6

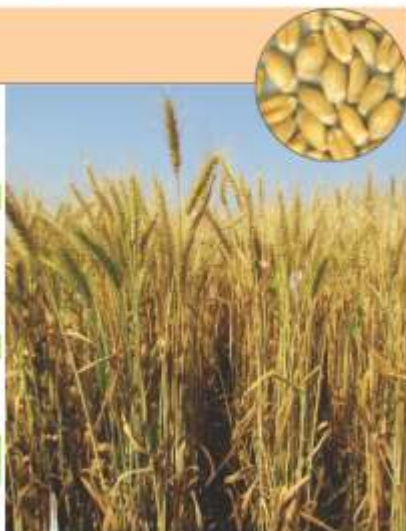
Quality parameters

Chapati score (max 10)	7.69
Bread loaf volume (cc)	558
Biscuit spread factor (max 10)	6.62

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	41.0
Grain yield under very late sown	33.0



Special Features: HD 2932 has shown wider adaptability across the zones with high yield potential. It has APR against brown and black rusts and therefore likely to be durable. This genotype has more number of grains and high thousand grain weight even under late sowings, thus indicating to soil temperature stress. It also high iron and zinc and good chapati making quality.

RAJ 4083

Zone: Peninsular Zone

Production condition: Late sown, Irrigated

Parentage	PBW 343/UP 2442//WR 258/UP 2425
Year of release	2007
Notification No.	1178(E), 20-07-2007
Developed by	RARS, Durgapura

Morphological features

Plant height (cm)	73
Days to maturity (days)	98
1000-grain weight (g)	38.3
Average yield (q/ha)	41.5
Potential yield (q/ha)	56.6

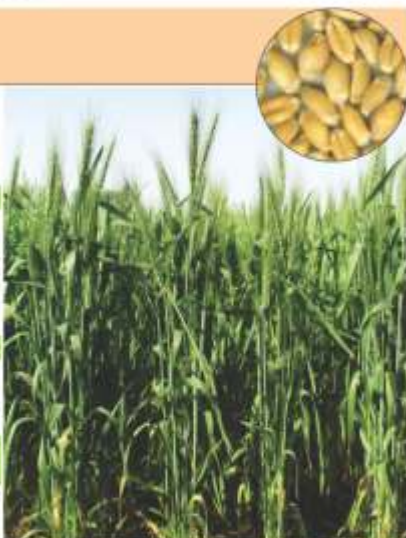
Quality parameters

Protein content (%)	12.6
Grain appearance	6.3
Hectolitre weight (g)	79.8

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	43.9
Grain yield under very Late sown	37.9



Special Features: Raj 4083 has shown potential for high yield and stability under late and very late conditions. The genotype possesses resistance to black and brown rusts, has high grain weight and early maturity to thrive well under warmer areas of India.

HD 2833

Zone: Peninsular Zone

Production condition: Late sown, Irrigated

Parentage	PBW226/HW1042//HD 2285
Year of release	2006
Notification No.	599(E), 25-04-2006
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	77
Days to maturity (days)	95
1000-grain weight (g)	40
Average yield (q/ha)	38.9
Potential yield (q/ha)	59.3

Quality parameters

Hectolitre weight (g)	81
Protein content (%)	13.3
Sedimentation value	44

Agronomic practices-

Grain yield (q/ha) under late/ very late planting

Grain yield under late sown	34.5
Grain yield under very late sown	31.2

Special Features: HD 2833 has shown wider adaptability and stability in yield under late and very late conditions. This genotype has shown APR to stem, leaf and yellow rusts and tolerance to leaf blight. This variety has good grain appearance and high hectolitre weight for better market price.



DBW 93

Zone: Peninsular Zone

Production condition: Timely sown, Restricted irrigated

Parentage	WHEAR/TUKURU//WHEAR
Year of release	2015
Notification No.	268 (E), 07-05-2015
Developed by	IIWBR, Karnal

Morphological features

Plant height (cm)	63
Days to maturity (days)	101
1000-grain weight (g)	37
Average yield (q/ha)	29.3
Potential yield (q/ha)	39.0

Quality parameters

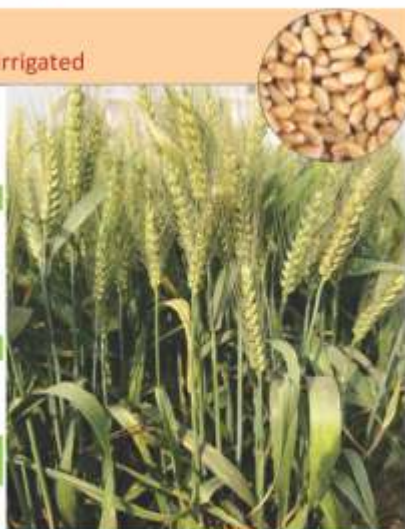
Chapati score (max 10)	7.69
Bread loaf volume (cc)	577
Biscuit spread factor (max 10)	6.24

Agronomic practices-

Grain yield (q/ha) under different irrigation levels

Grain yield under one irrigation	22.6
Grain yield under two irrigations	28.4

Special Features: DBW 93 is a high yielding genotype suitable for rainfed and restricted area of peninsular India. This variety has shown wider adaptability in terms of moisture regimes as well as nitrogen levels. It is good for chapati, bread and biscuit making quality alongwith better nutritional components.



NIAW 1415

Zone: Peninsular Zone

Production condition: Timely sown, Rainfed/Restricted irrigated

Parentage	GW 9506/PRL//PRL
Year of release	2011
Notification No.	622(E), 25-03-2011
Developed by	ARS, Nipad

Morphological features

Plant height (cm)	69(RF), 73(RI)
Days to maturity (days)	122 (RF), 103(RI)
1000-grain weight (g)	35.1(RF), 35(RI)
Average yield (q/ha)	19.5 (RF), 31.1(I)
Potential yield (q/ha)	36.2 (RF), 38.2 (I)

Quality parameters

Chapati score (max 10)	8.06(RF), 8.03(RI)
Bread loaf volume (cc)	573(RF), 572(RI)
Biscuit spread factor (max 10)	7.35 (RF), 7.09(RI)

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose (RF)

Grain yield (q/ha) under different irrigation levels (RI)

Grain yield under N60	22.9(RF),	Grain yield under one irrigation	33.87(RI)
Grain yield under N80	25.5(RF),	Grain yield under two irrigations	35.0(RI)

Special Features: NIAW 1415 has good adaptability to rainfed as well as restricted irrigation condition and showed responsiveness to higher level of nitrogen. The variety has shown high degree of resistance to leaf and brown rusts, has good grain appearance and rich in iron contents.



HD 2987

Zone: Peninsular Zone

Production condition: Timely sown, Rainfed/Restricted irrigated

Parentage	HI 1011/HD 2348//MENDOS//IWP 72/DL 153-2
Year of release	2011
Notification No.	632(E), 25-03-2011
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	56(RF), 64(RI)
Days to maturity (days)	96(RF), 103(RI)
1000-grain weight (g)	40(RF), 38.0(RI)
Average yield (q/ha)	17.5 (RF), 31.5 (RI)
Potential yield (q/ha)	32.2(RF), 38.6 (RI)

Quality parameters

Chapati score (max 10)	8.14(RF), 8.02(RI)
Bread loaf volume (cc)	583(RF), 580(RI)
Biscuit spread factor (max 10)	5.76(RF), 6.19(RI)

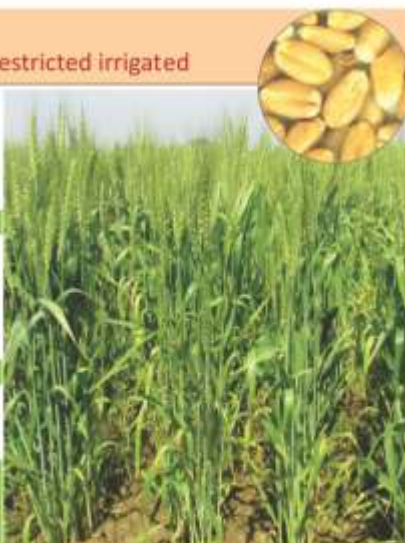
Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose (RF)

Grain yield (q/ha) under different irrigation levels (RI)

Grain yield under N60	19.9(RF)	Grain yield under one irrigation	26.0(RI)
Grain yield under N80	20.1(RF)	Grain yield under two irrigations	29.0(RI)

Special Features: HD 2987 has shown high yield potential under rainfed as well as restricted irrigation conditions over years and locations. This variety has also shown APR to brown and black rust. This variety has high iron, copper and manganese and high score for chapati quality.



HD 2781

Zone: Peninsular Zone

Production condition: Timely sown, Rainfed

Parentage	BOW /C 306 //C 591/HW 2004
Year of release	2002
Notification No.	937(E), 04-09-2002
Developed by	IARI, New Delhi

Morphological features

Plant height (cm)	63
Days to maturity (days)	110
1000-grain weight (g)	45
Average yield (q/ha)	15.8
Potential yield (q/ha)	23.6

Quality parameters

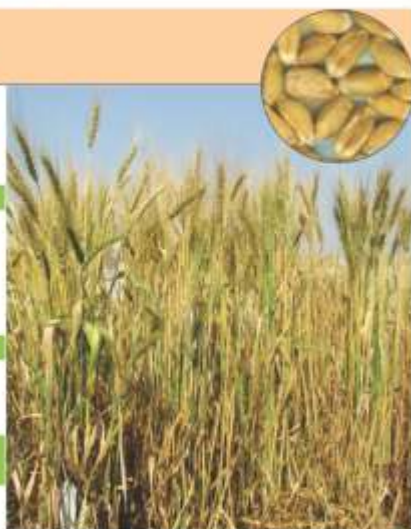
Protein content (%)	12.0
Grain appearance score (max 10)	6.4
Sedimentation value	49.3

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N60	12.2
Grain yield under N80	12.5

Special Features: HD 2781 has shown very good adaptation under higher temperature and moisture stress conditions. The genotype has resistance to leaf and stem rust. This variety also has got 'stay green' habit and slow senescence.



UAS 446 (Durum)

Zone: Peninsular Zone

Production condition: Timely sown, Rainfed

Parentage	DWR 185/ DWR 206/ UAS 419
Year of release	2015
Notification No.	1228 (E), 07-05-2015
Developed by	UAS, Dharwad

Morphological features

Plant height (cm)	66
Days to maturity (days)	96
1000-grain weight (g)	44
Average yield (q/ha)	18.3
Potential yield (q/ha)	24.4

Quality attributes

Yellow Pigment	5.70
Protein content (%)	13.9
Hectolitre weight (kg/hl)	82.6

Agronomical features-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N 60	14.3
Grain yield under N 80	14.8

Special Features: UAS 446 is a high yielding durum wheat genotype suitable for rainfed condition and responded well to different doses of fertilizers. It is resistant to black & brown rusts.



Southern Hill Zone

Varieties at a glance

Condition	Variety
Timely sown, restricted irrigated	Bread Wheat : HW 5216, COW (W)-1, HW 2044 Dicoccum Wheat : HW 1098

Package of Practices

Conditions	Sowing time	Seed rate	Fertilizer doses and time of application
Timely sown, Restricted irrigated	2 nd week of November to 2 nd week of December	100 kg/ha	90:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)

HW 5216

Zone: Southern Hill Zone

Production condition: Timely sown, Restricted irrigated

Parentage	PBW 343//HW 3083
Year of release	2013
Notification No.	952(E), 10-04-2013
Developed by	IARI-RS, Wellington

Morphological features

Plant height (cm)	86
Days to maturity (days)	105
1000-grain weight (g)	43.7
Average yield (q/ha)	45.6
Potential yield (q/ha)	62.4

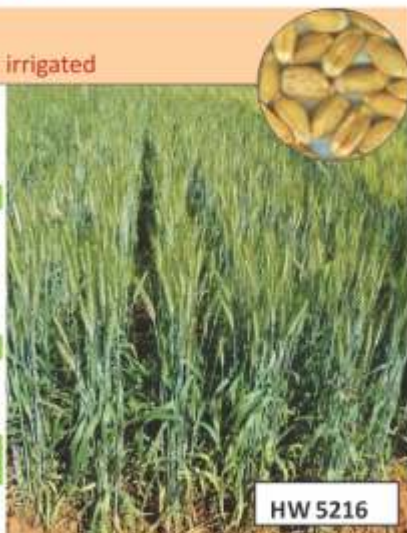
Quality parameters

Chapati score (max 10)	7.60
Bread loaf volume (cc)	555
Biscuit spread factor (max 10)	7.49

Agronomic practices-

Grain yield (q/ha) under different irrigations levels

Grain yield under one irrigation	30.8
Grain yield under two irrigation	33.9



Special Features: HW 5216 has good yield potential under varied irrigation level, has high degree of seedling resistance to stem, leaf and yellow rusts against all the prevalent races. This variety has good grain appearance, high test weight and better grain quality.

COW (W)-1

Zone: Southern Hill Zone

Production condition: Timely sown, Restricted irrigated

Parentage	HD2646/HW2002A/CPAN3057
Year of release	2006
Notification No.	599(E), 25-04-2006
Developed by	IARI-RS, Wellington

Morphological features

Plant height (cm)	85
Days to maturity (days)	104
1000-grain weight (g)	37
Average yield (q/ha)	23.6
Potential yield (q/ha)	36.4

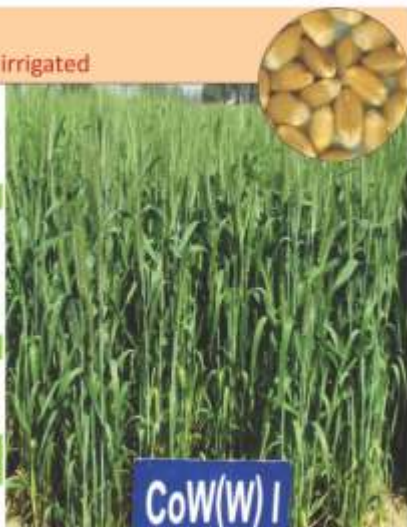
Quality parameters

Chapati score (max 10)	8.14
Sedimentation value	51
Hectolitre weight (g)	78

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	-
Grain yield under late sown	-



Special Features: COW (W)-1 is a medium dwarf high yielding disease resistant genotype for irrigated and rainfed areas. It has good chapati making quality and high sedimentation value. 396

HW 2044

Zone: Southern Hill Zone

Production condition: Timely sown, Rainfed/Irrigated

Parentage	HD 226*5/SUNSTAR*6/C-80-1
Year of release	2000
Notification No.	340(E), 03-04-2000
Developed by	IARI-RS, Wellington

Morphological features

Plant height (cm)	92
Days to maturity (days)	92
1000-grain weight (g)	40.0
Average yield (q/ha)	54.0
Potential yield (q/ha)	58.7

Quality parameters

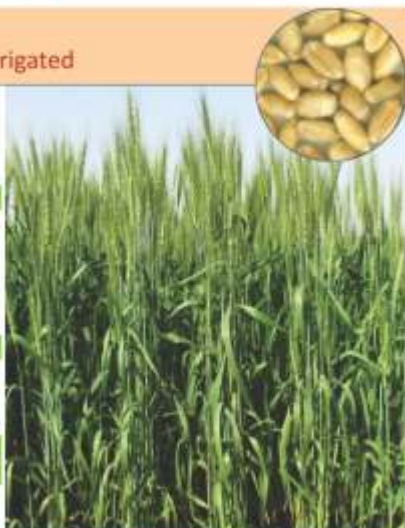
Chapati score (max 10)	7.75
Bread loaf volume (cc)	313
Biscuit spread factor (max 10)	7.43

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	-
Grain yield under late sown	-

Special Features: HW 2044 has high yield potential and recommended for cultivation in irrigated as well rainfed areas under southern hill zone. It is a medium dwarf genotype having bold grain and good chapati Quality parameters.



HW 1098 (Dicoccum)

Zone: Southern Hill Zone

Production condition: Timely sown, Irrigated

Parentage	NP 201 (Mutant developed through 20 Kr irradiation)
Year of release	2015
Notification No.	268(E), 28-01-2015
Developed by	IARI-RS, Wellington

Morphological features

Plant height (cm)	85
Days to maturity (days)	106
1000-grain weight (g)	40.3
Average yield (q/ha)	45.5
Potential yield (q/ha)	59.0

Quality parameters

Protein content (%)	13.5
Sedimentation value	24
Yellow pigment (ppm)	3.39

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	40.5
Grain yield under late sown	32.7

Special Features: Dicoccum variety HW 1098 has high yield potential and showed wider adaptability for planting under normal and late sowing conditions. This is a semi dwarf variety that has required level of resistance, medium bold grains, early maturing and good protein content.



Marginal Areas

Saline/Alkaline Soils

Varieties at a glance

Condition	Variety
Timely sown, Irrigated-Saline/Alkaline soils	KRL 210, KRL 213, KRL 19

Package of Practices

Conditions	Sowing time	Seed rate	Fertilizer doses and time of application
Timely sown, Irrigated	2 nd - 3 rd week of November	100 kg/ha	120:60:40 kg NPK /ha (1/3 N at sowing and remaining in two equal splits at first and second irrigation)

KRL 210

Zone: Northern Plains Zone

Production condition: Timely sown, Irrigated-Saline soils

Parentage	PBW 65/2*PASTOR
Year of release	2012
Notification No.	456(E), 16-03-2012
Developed by	CSSRI, Karnal

Morphological features

Plant height (cm)	99
Days to maturity (days)	143
1000-grain weight (g)	40.1
Average yield (q/ha)	33.7
Potential yield (q/ha)	49.3

Quality parameters

Protein content (%)	10.9
Sedimentation value	39.3
Hectolitre weight (g)	77.1

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N120	32.2
Grain yield under N150	34.8



Special Features: KRL 210 has shown high yielding ability alongwith salt tolerance over years and locations. This variety is lodging resistance, possessing resistance to yellow and brown rust alongwith good quality parameters.

KRL 213

Zone: Northern Plains Zone

Production condition: Timely sown, Irrigated-Saline soils

Parentage	CNDO/RI43/ENTE/MEXL -2/3/Ae. Sq.
	TAUSO/4/WEAVER/5/28KAUZ
Year of release	2011
Notification No.	2326(E), 10-10-2011
Developed by	CSSRI, Karnal

Morphological features

Plant height (cm)	97
Days to maturity (days)	145
1000-grain weight (g)	36.1
Average yield (q/ha)	32.5
Potential yield (q/ha)	43.9

Quality parameters

Protein (%)	10.8
Sedimentation value	28.7
Hectolitre weight (g)	77.2

Agronomic practices-

Grain yield (q/ha) under enhanced fertilizer dose

Grain yield under N120	35.0
Grain yield under N150	36.3



Special Features: Wheat variety KRL 213 has high yield potential under harsh environments and thus recommended for cultivation in salt affected areas under north India (NWPZ & NEPZ) This genotype has also shown yield gain with enhanced dose of nitrogen. It is resistant to yellow and brown rusts, tolerant to leaf blight and Karnal bunt.

KRL 19

Zone: All Zones

Production condition: Timely sown, Irrigated - Alkaline / Saline soils

Parentage	PBW 255/KRL 1-4
Year of release	1999
Notification No.	340(E), 03-04-2000
Developed by	CSSRI, Karnal

Morphological features

Plant height (cm)	70
Days to maturity (days)	130
1000-grain weight (g)	38
Average yield (q/ha)	30.4
Potential yield (q/ha)	46.4

Quality parameters

Protein content (%)	13.09
Hardens index	82.6
Sedimentation value	45

Agronomic practices-

Grain yield (q/ha) under normal / late planting

Grain yield under normal sown	27.6
Grain yield under late sown	16.8

Special Features: KRL 19 has been found suitable for salt affective areas. This genotype was responsive to fertilizers and tolerant to lodging. It has good grain appearance, high protein and hectolitre weight.



Crop Management Practices

Weed management: Wheat is infested with both grasses and broad-leaved weeds and for realizing full yield potential of wheat crop, proper weed control is essential. Among various methods of weed control, chemical method is the most effective and depending on weed flora following herbicides can be applied:

Weed flora	Herbicide	Dose g/ha	Time of application DAS
Grasses	Pendimethalin (Stomp 30EC)	1000-1500 (3333-4950)	Pre-emergence 1-3 DAS
	Clodinafop (Topik 15WP)	60 (400)	Post emergence 30-35 DAS
	Fenoxaprop-ethyl (Puma Super 10EC)	100-120 (1000-1200)	Post emergence 30-35 DAS
	Pinoxaden (Axial 5 EC)	35-40 (700-800)	Post emergence 30-35 DAS
	Sulfosulfuron (Leader 75 WG)	33.3 (25)	Post emergence 30-35 DAS
	Isoproturon (Arelon 75 WP)	1000 (1333)	Post emergence 30-35 DAS
Broad-leaved weeds	Pendimethalin (Stomp 30EC)	1000-1500 (3333-4950)	Pre-emergence 1-3 DAS
	2,4-D-E (Weed war 38 EC)	500(1315)	Post emergence 30-35 DAS
	Metsulfuron (Algrip 20 WP) P)	4 (20)	Post emergence 30-35 DAS
	Carfentrazone (Affinity 50 WDG)	20 (50)	Post emergence 30-35 DAS
Both grassy and broad-leaved weeds	Sulfosulfuron (Leader 75 WG)	33.3 (25)	Post emergence 30-35 DAS
	Pendimethalin (Stomp 30EC)	1000-1500 (3333-4950)	Pre-emergence 1-3 DAS
	Isoproturon (Arelon 75WP)+2,4-D E	750 + 500 (1000+1315)	Post emergence 30-35 DAS
	Sulfosulfuron+ metsulfuron (Leader 75WG+ Algrip 20WP)	25 +2 (33.3)	Post emergence 30-35 DAS

- In areas having problem of isoproturon resistant *P. minor*, use sulfosulfuron, or clodinafop or fenoxaprop or pendimethalin or pinoxaden. Sulfosulfuron and pendimethalin are effective against both grassy and non-grassy weeds whereas, clodinafop, fenoxaprop, and pinoxaden are specific to grasses. Areas infested with multiple herbicide resistant *P. minor* (resistant to isoproturon, clodinafop and sulfosulfuron) use of pre-emergence pendimethalin is suggested.

- Combination of herbicides should be applied for the control of complex weed flora. Sulfosulfuron + metsulfuron and isoproturon with 2,4-D or metsulfuron can be used as tank mixture. Grass herbicides (clodinafop, fenoxaprop, and pinoxaden) should not be tank mixed with either 2,4-D or metsulfuron and to avoid antagonism the grass and broad-leaved herbicides should be applied sequentially.

Water Management : Four to six irrigations are enough for raising a good wheat crop. However, crown root initiation and flowering stages are the most critical to moisture stress. Depending on water availability, irrigation should be applied. In case of very light soils as well as in areas having undulating topography, sprinkler method of irrigation should be used. Also in light soils and in area with water scarcity, bed planting techniques can be adopted as it saves 30% water. Further, laser land leveling must be popularized on large scale as it will help in improving the input use efficiency of both water and nitrogen on fertilizer.

Disease/pest management

Yellow rust (NHZ & NWPZ) : Stripe rust (yellow rust) is very important in NHZ and NWPZ and therefore resistant varieties should only be grown. It is generally observed that early infection of stripe rust starts in wheat fields with the poplar trees wherever it is early sown (i.e. October). Hence, strict watch is needed for such fields. Under situation of heavy inoculum load of stripe rust due to favourable climate conditions, chemical sprays with propiconazole (Tilt 25 EC @ 0.1 per cent), is recommended for disease management.

Leaf rust (All zones) : Leaf rusts are common across the wheat growing zones of the country and cultivating resistant variety is the most suitable approach to avoid losses. There are leaf rust tolerant genotypes available for each zone and production condition. In case of unavoidable situation, chemical spray option be exercised.

Stem rust (CZ & PZ) : Cultivation of rust resistant varieties is the best practice to avoid stem rust incidence.

Loose smut : Seed treatment with Carboxin (75 WP @ 2.5 gm/kg seed) or Carbendazim (50 WP @ 2.5 gm/kg seed) or Tebuconazole (2DS @ 1.25 gm/kg seed) or a combination of a reduced dosage of Carboxin (75 WP @ 1.25 gm/kg seed) and a bioagent fungus *Trichoderma viride* (@ 4 gm/kg seed) is recommended.

Karnal bunt : Karnal bunt control is required wherever crop is grown for seed or the produce for export purposes. For producing KB free wheat, farmers are advised to grow KB resistant varieties recommended for the respective area. Growing durum wheat for 2-3 years, reduces Karnal bunt pathogen load in the soil. Zero tillage helps in reducing Karnal bunt incidence. One spray of Propiconazole 25EC (Tilt 25 EC) @ 0.1 per cent be given after heading to control the disease.

Powdery mildew : For powdery mildew control, one spray of propiconazole (Tilt 25 EC) @ 0.1 % at ear head emergence or appearance of disease (whichever is earlier) is recommended for powdery mildew prone areas.

Foliar blight : Foliar blight is the main wheat crop health problem in NEPZ. For effective management of the diseases, cultivation of recommended varieties, like HD 2985, HI 1563, DBW 39, CBW 38, NW 1014, NW 2036, HD 2733, DBW 14, HD 2888, K 0307, K 1006 and NW 5054 should be encouraged.

Flag smut : Flag smut disease also poses problems in isolated fields in Punjab, Haryana, Rajasthan and some other parts of NWPZ. Disease management measures taken for the control of loose smut disease (as discussed above), prove to be effective against flag smut too. Hence, seed treatment with carboxin or tebuconazole may be followed in fields with flag smut recurrence history.

Termite : In the termite prone areas, seed treatment with chlorpyrifos @ 0.9g *a.i.*/kg seed, be taken up for its management. Seed treatment with thiamethoxam 70WS (Cruiser 70WS) @ 0.7 g *a.i.*/kg seed or Fipronil (Regent 5FS @ 0.3 g *a.i.*/kg seed) is also very effective. In the standing crop, broadcasting of the insecticide treated soil using chlorpyrifos @ 3 litre mixed in 50 kg soil be used for one hectare field. Crop planted under FIRBS is more prone to termite attack in the termite-prone areas, while zero tillage shows less termite damage. Hence, attention should be given to crop planted under FIRBS.

Aphids : For the management of aphids, foliar spray of imidacloprid 200SL @20g *a.i.*/ha on border rows at the start of the aphid colonization be given. This will help in protection of the bioagent insect, the lady bird beetle inside the field which feeds on aphids.

Pink stem borer : The incidence of pink stem borer has been observed more in fields of rice-wheat cropping system where wheat is sown in zero tillage fields. For its management, foliar spray of quinalphos(Ecalux) 800 ml /acre is recommended as soon as pink stem borer is seen. Irrigation also helps in reducing the pink stem borer damage.

Ear cockle : Ear cockle is an important disease in eastern parts of India, hence proper precautions be taken, especially in eastern U.P., Bihar, Rajasthan and Jharkhand. Wider publicity should be given by extension agencies on the use of gall-free seed, well before the sowings. Farmers should adopt floatation technique for the separation of galls from the infested seed lots. The infested seed lot should be floated in 2 percent brine solution for this purpose. The galls will float on the surface which should be separated and destroyed away from the field by burning. The seed should be thoroughly washed to remove the salt solution before sowing.



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